



OPEN Explainable artificial intelligence for predictive modeling of student stress in higher education

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Student stress in higher education remains a pervasive problem, yet many institutions lack affordable, scalable, and interpretable tools for its detection and management. Existing methods frequently depend on costly physiological sensors and opaque machine learning models, limiting their applicability in resource-constrained settings. The objective of this research is to develop a cost-effective, survey-based stress classification model using multiple machine learning algorithms and eXplainable Artificial Intelligence (XAI) to support transparent and actionable decision-making in educational environments. Drawing on a dataset of university students, the research applies a supervised machine learning pipeline to classify stress levels and identify key contributing variables. Six classification algorithms—Logistic Regression, Support Vector Machine (SVM), Decision Tree, Random Forest, Gradient Boosting, and XGBoost—were employed and optimized using grid search and cross-validation for hyperparameter tuning. Evaluation metrics included precision, recall, F1-score, and overall accuracy. The Random Forest model achieved the highest classification accuracy of 0.89, followed by XGBoost at 0.87, Gradient Boosting at 0.85, Decision Tree at 0.83, SVM at 0.82, and Logistic Regression at 0.81. SHAP (SHapley Additive exPlanations) analysis was conducted to interpret model predictions and rank feature importance. The analysis revealed five principal predictors: blood pressure, perceived safety, sleep quality, teacher-student relationship, and participation in extracurricular activities. Results demonstrate that both physiological indicators and psychosocial conditions contribute meaningfully to stress prediction. The study concludes that institutional interventions targeting health monitoring, campus safety, behavioral support, relational pedagogy, and extracurricular engagement can effectively mitigate student stress. These findings provide an empirical foundation for the development of integrated policies in higher education aimed at promoting student well-being.

Keywords Machine learning, Educational innovation, Student stress factors, Higher education, Game theory, Stress intervention strategies, Explainable AI in education, Student mental health analytics

In educational settings, the significance of understanding student stress factors cannot be overstated. Stress among students has emerged as a prevalent issue, profoundly influencing their academic performance, mental health, and overall well-being¹. The roots of this stress are diverse, from academic challenges, and social pressures, to the multiple demands of student life². In an era increasingly guided by data-driven insights, the application of advanced analytical techniques to identify and understand these stress factors is not only innovative but crucial. By delving into the complex relations of variables that contribute to student stress³, researchers and educators can gain a better understanding of the issue⁴. This, in turn, is pivotal for developing effective strategies to support students in their educational journeys and beyond⁵.

Artificial intelligence method specifically Decision trees (DTs) (and similar algorithms) stand as a fundamental method in the realm of classification algorithms, renowned for their simplicity and effectiveness⁶. At their core, DTs partition data into subsets based on a series of simple, decision-like questions. This process of splitting the data is based on the features within the dataset. Each split aims to create groups that are as homogeneous as

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possible in terms of the target variable. This attribute of DTs (and similar algorithms) makes them particularly suitable for classifying complex datasets⁷. The intuitive nature of DTs, where each decision point resembles a 'branch' in a tree, allows for easy interpretation and visualization of how different features contribute to the overall classification⁸. In analyzing student stress factors, DTs (and similar algorithms) can effectively discern patterns and relationships within intricate datasets, providing valuable insights into the various elements that influence student well-being⁹.

Grounded in the principles of game theory¹⁰, SHapley Additive exPlanations (SHAP) represent a noteworthy advancement in the interpretation of machine learning (ML) models. This approach provides a comprehensive framework for attributing the output of a model to its input features, thereby assigning an importance value to each feature for a particular prediction¹¹. SHAP values facilitate a deeper understanding of how a model makes decisions by quantifying the contribution of each feature. This aspect of SHAP is particularly relevant in the context of high-dimensional data, where numerous variables interact in complex ways to influence the model's predictions. By applying SHAP, researchers can unravel the opaque decision-making process of complex and sophisticated models, shedding light on the specific features that drive predictions. This level of interpretability is invaluable when exploring student stress factors, where understanding the impact of each contributing element is essential for drawing meaningful conclusions.

Theoretical framework and current-state-of-the-art

Stress factors in students and study methods

In the present and future of education^{12,13}, identifying stress factors in students that may affect their mental health and ability to learn is important to implement effective support strategies and promote a healthy educational environment. Researchers such as da Silva Ezequiel et al.¹⁴ conducted a 30-month longitudinal study to assess factors associated with motivation in students, applying the MSSF-Medical Student Stress Factor scale, finding factors such as learning approach, hours spent studying, gender, stressors, studying just before exams and sleep problems were predictors of different dimensions of motivation. Also, Damiano et al.¹⁵ found that the major stressors were lengthy content, lack of time to study, sleep deprivation, excessive self-pressure to get good grades and lack of free time, also finding that the factors are related to academic performance and the learning environment and should be targeted in future interventions. Moldt et al.¹⁶ applied the Perceived¹⁷ Stress Questionnaire and Chatbots as personal digital assistants to measure medical students' stress levels in everyday conversations and identified students' acceptance of chatbots as interlocutors. The studies shed light on the location of prevention programmes and it is important to be cautious about generalising the scale of student stressors¹⁸. Working together to implement stress prevention and stress management strategies can help create a healthier, more inclusive and equitable educational community for all students.

Study types to analyse stress have integrated diverse methodologies. For example, Saleh et al.¹⁹ conducted regression analyses to evaluate a model of stress vulnerability in French university students and found that life satisfaction, self-esteem, optimism, self-efficacy and psychological distress were the most important predictors of stress. Meanwhile, Bayrak et al.²⁰ analysed differences in perceived stressors and stress symptoms, in a mediation model, locating that the effect of anxious attachment on perceived stressors and stress symptoms was partially mediated by self-concept and coping styles, where the level of anxious attachment is an important factor in explaining perceived stress and stress-related variables. Mohd & Yahya²¹ used two data mining techniques: Logistic Regression and Artificial Neural Network, to compare the performance of different data mining algorithms for predicting depression among students, showing that the Artificial Neural Network technique predicts depression more accurately from these two classification models. In distance environments, Marsella & Citrayasa²² analysed stress factors in online learning, finding that the major stress factor was the lack or absence of interaction between students and with teachers, as well as the abundant number of homework assignments given by teachers. The different methodologies provide strengths for exploring the complexities of student experiences and encourage substantial consideration of them in prevention programmes.

Educational innovation and student stress

The role of educational innovation in adopting student well-being, alleviating stress, and improving academic success has become increasingly prominent in contemporary education research. Innovative teaching practices and methodologies have shown potential to transform the educational landscape by creating supportive learning environments that promote mental health and enhance academic outcomes.

Research highlights the significant impact of integrating novel teaching approaches on students' emotional and psychological well-being. For instance, the use of mindfulness-based interventions, as demonstrated by Strout et al. (2024)²³, has been effective in reducing stress and enhancing well-being among nursing students. By incorporating evidence-based mindfulness practices within the curriculum, educational institutions can provide students with tools to manage stress effectively, which in turn positively influences their academic performance and satisfaction with their learning experience.

Similarly, experiential and gamified learning approaches, such as escape rooms, have proven to be impactful in both reducing anxiety and increasing engagement. Dogu et al. (2025)²⁴ observed that nursing students who participated in escape room activities not only exhibited lower stress levels but also demonstrated enhanced knowledge retention and motivation. These findings suggest that interactive and immersive methods can improve both cognitive and emotional aspects of learning.

Technological innovations, including digital concept mapping and blended learning, further highlight the potential of modern educational tools to address stress and promote academic success. Tang and Tang (2024)²⁵ found that the use of digital concept mapping significantly reduced learning anxiety while simultaneously increasing students' motivation and self-regulated learning behaviors. Similarly, Salim et al. (2024)²⁶ noted that

blended learning environments fostered students' self-efficacy and self-actualization, essential components of emotional and academic resilience.

The integration of sustainability and well-being principles into educational frameworks also plays a critical role. Ramírez-Montoya et al. (2024)²⁷ emphasize the importance of aligning educational strategies with the goals of Education 5.0, which uses technological resources to maximize human well-being. This alignment not only enhances academic outcomes but also equips students with the skills and mindset needed for long-term success and societal contribution.

Lastly, using a sense of belonging and community in educational settings has been shown to mitigate feelings of isolation and stress among students. Edwards and Hardie (2024)²⁸ demonstrated that institution-wide online events designed to build community and engagement significantly improved students' sense of belonging, which correlates with higher academic achievement and emotional well-being. This outlines the necessity for educational institutions to create inclusive and supportive environments that address both psychological and social needs.

Overall, the evidence indicates that educational innovation, when thoughtfully implemented, can significantly reduce student stress, enhance well-being, and improve academic success. These findings advocate for a paradigm shift in education, where student-centered, innovative, and inclusive approaches are prioritized to create sustainable and supportive learning environments.

ML methods in stress detection with explainable artificial intelligence perspective

The detection and analysis of stress in students have been greatly enhanced by the advent of ML algorithms²⁹, which provide novel avenues for non-invasive, efficient, and accurate stress monitoring. Nazeer et al.³⁰ introduced STRESS-CARE, an innovative sensor system leveraging Galvanic Skin Response (GSR) and ML techniques, to overcome the rigidity and noise interference challenges of traditional methods such as ECG and body movement analysis. Utilizing an XG Boost classifier, this approach not only excels in detecting stress with higher accuracy but also adapts to different environmental conditions, marking a significant advancement in wearable medical technology for stress management.

Parallel to this, Naegelin et al.³¹ demonstrated the effectiveness of ML models in stress detection using unobtrusive multimodal data collected from a simulated group office environment. Their methodology, which hinges on mouse, keyboard, and cardiac data, underscored that behavioral data might be more indicative of stress levels in office settings than cardiac data, with SHapley Additive exPlanations (SHAP) providing interpretability to the machine learning predictions. Such insights are vital for the development of just-in-time adaptive interventions aimed at stress management in real-world conditions.

Additionally, Gonzalez-Carabarin et al.³² presented a specific stress evaluation utilizing EEG and ECG biomarkers, processed through semi-supervised machine learning techniques to cater to individual variability in stress responses. The stress metrics derived from this study enable personalized diagnostics, offering a foundation for real-time monitoring and prevention of chronic health issues related to stress. Panicker and Gayathri³³ surveyed the role of machine learning in physiology-based stress detection systems, highlighting the links between biological features and mental stress and the potential for future research. This comprehensive review points to the importance of diverse data collection methods and machine learning algorithms in developing robust stress detection systems.

The research by Ahuja and Banga³⁴ focuses specifically on the mental stress of university students, employing machine learning algorithms like SVM to analyze stress levels during critical periods such as exams and extensive internet use. The high accuracy achieved by their models underlines the potential of machine learning in identifying stressors within educational settings, providing valuable data for interventions. Furthermore, Vos et al.^{35,36} explored the use of ensemble machine learning models to improve the generalization of stress prediction using wearable devices, addressing the challenge of small datasets by proposing a synthesis method for dataset enhancement. Their approach not only demonstrates a substantial increase in predictive performance but also contributes to reproducible research by making their datasets and code publicly accessible.

The study by Ratul et al.³⁷ emphasizes the exacerbated psychological and social stress among university students due to the COVID-19 pandemic, with their machine learning-based model showcasing high effectiveness in early stress detection through an online survey. This underscores the growing relevance of machine learning in mental health interventions, especially in response to global crises that significantly impact student well-being. Daza et al.³⁸ conducted a systematic review focusing on machine learning's role in predicting stress and anxiety in college students, emphasizing the predominance of SVM and Logistic Regression algorithms, executed predominantly in Python, and gauging model effectiveness using precision and accuracy as key metrics. Similarly, Bhatnagar et al.³⁹ applied a variety of ML algorithms, including Naïve Bayes and Random Forest, to classify anxiety levels among engineering students, achieving significant accuracy.

In an era where educational settings are rapidly digitizing, Sergio et al.⁴⁰ proposed an eHealth system that harnesses IoT devices for the continuous monitoring of vital signs, with machine learning models identifying patterns indicative of stress and anxiety. This initiative exemplifies the integration of technology in promoting student well-being. Conversely, Shah et al.⁴¹ reviewed the efficacy of ML in neuropsychological detection and prediction, revealing the robustness of SVMs and the proficiency of CNNs in neuroimaging, thereby underlining the versatility of ML algorithms across diverse applications in mental health. Mittal et al.⁴² reviewed ML applications in stress management within workplaces and educational environments, highlighting the urgent need for technologies capable of early stress detection to mitigate potential health crises. This work outlines the significant impact of the COVID-19 pandemic on mental health and the potential of ML in addressing the ensuing challenges. Rayan and Alanazi⁴³ ventured into forecasting mental well-being through ML, collating a comprehensive dataset that spans behavioral, psychological, and environmental factors, processed using advanced ML models to provide timely mental health insights. Singh and Kumar⁴⁴ offered a systematic review

of ML algorithms' capacity to identify stress, anxiety, and depression (SAD) through various data forms like questionnaires and biometric measures. Their findings underscore the profound capability of ML in recognizing and monitoring SAD symptoms.

ML has significantly advanced the identification and understanding of student stress, with studies demonstrating the efficacy of various algorithms and biometric measures to detect stress-related symptoms. Innovations range from the STRESS-CARE sensor system for physiological monitoring to multimodal data analyses in simulated environments, emphasizing the importance of behavioral over cardiac data in certain contexts. Ensemble models and eHealth systems integrating IoT devices have also been pivotal, enhancing data collection and model generalization. However, despite these technological strides, there is a consensus that further research is required, utilizing diverse ML methods to refine the accuracy and applicability of stress detection among students, ensuring that interventions are both timely and effective in mitigating the impact of stress on student well-being.

Research gaps, necessity, and importance of current research work

Table 1 provides a comprehensive overview of recent research studies focusing on stress detection in students using various physiological, behavioral, and computational approaches. These studies demonstrate the diversity of methodologies and technologies applied to identify and classify stress levels, reflecting the evolving sophistication in this field. For example, several studies^{45,46} use electroencephalogram (EEG) signals for stress detection, employing machine learning algorithms such as LightGBM and CNN to achieve high accuracy rates in classification tasks. Similarly, other research has focused on the integration of physiological signals, such as ECG, EMG, and GSR, to detect stress through multi-level classification, as seen in the work of Pourmohammadi and Maleki⁴⁷ and Tiwari and Agarwal⁴⁸. Additionally, studies like Zhu et al.⁴⁹ and Oryngozha et al.⁵⁰ explore non-physiological data sources, including textual comments and social media posts, utilizing natural language processing and deep learning techniques to detect stress indicators in academic settings. The integration of wearable technologies, as demonstrated by Pérez et al.⁵¹, and mobile sensors, as in Luo et al.⁵², highlights the potential for scalable, real-time stress monitoring. Furthermore, studies such as those by Figueroa et al.⁵³ and Morales-Rodríguez et al.⁵⁴ emphasize interventions and coping strategies alongside stress detection, providing a holistic approach to managing student stress.

Despite the usage of advanced technologies, which can often be prohibitively expensive and may not be feasible for universities operating under constrained budgets, there remains a critical need for low-cost programs and setups to address student stress factors. Universities worldwide typically allocate their budgets to core expenditures such as laboratory equipment acquisition, academic staff salaries, and essential operational costs, leaving initiatives for student well-being to operate with limited financial resources. Developing affordable and accessible technologies for stress detection is essential to ensure such initiatives do not disrupt the overall financial distribution of the institution. Additionally, the type of input data required for stress detection models can also be costly, and its collection presents significant challenges. This emphasizes the need for minimal data-driven, cost-effective solutions that can augment stress detection processes in students. Furthermore, prioritizing hardware-free, straightforward procedures for stress detection can enable wider adoption across diverse university settings. Such advancements not only offer practical solutions for stress management but also provide valuable data and insights into key stress factors affecting students. These insights can guide student well-being administrators and student affairs officers in formulating targeted intervention policies, ultimately reducing the risk of student stress and advancing healthier academic environments.

One such similar initiative has been recently published by de Filippis & Foysal (2024)⁵⁵, however, the focus remained on the graphical presentation of student stress factors, and only using the Random Forest algorithm to present the importance of the specific factor. Consequently, this leads to a black box approach where the specific contribution of the decision variables does not seem to be well-explainable with the student stress index. Eventually, the classification fit performance of the Random Forest algorithm is also absent, therefore, this research requires further work specifically in the direction of applying various machine learning algorithms to select the best fit model of this classification problem, and eventually, based upon this, apply an eXplainable Artificial Intelligence (XAI) procedure like Shapley values are required to justify the intervention and policy development. Thus, in summary, there is a need of research to develop cost-effective only data-driven solutions are required, and additionally, there is a need to apply multiple machine learning algorithms to address this problem while considering explainable AI context.

This research addresses this critical gap in the field of student mental health analytics by proposing a cost-effective, hardware-independent framework for stress detection in higher education using supervised machine learning and XAI. In contrast to prior studies that often rely on expensive physiological sensors or black-box models, this study focuses on developing an interpretable, survey-based approach that remains accessible to institutions operating with limited financial and technical resources. By applying multiple machine learning algorithms and integrating SHAP, the research enables a transparent analysis of how specific psychological, physiological, social, environmental, and academic variables contribute to student stress. This methodological shift responds to the pressing need for scalable, evidence-based tools that allow decision-makers to implement targeted interventions without incurring prohibitive costs. The main objective of the research is to construct a data-driven stress classification model that prioritizes explainability and affordability, offering a practical and replicable solution for academic institutions seeking to enhance student well-being through informed policy-making and resource allocation. This contribution not only strengthens the operational feasibility of mental health analytics in education but also advances the integration of XAI into socio-educational contexts where accountability and interpretability are paramount.

Reference	Summary	Advantages	Disadvantages
Fernandez et al. (2024) ⁴⁵	This study investigates stress detection using EEG signals, focusing on the neural correlates of stress in students. Physiological data were recorded, and stress was induced through experimental setups. Four machine learning models—LightGBM, CNN, KNN, and SVM—were employed for classification. Results showed that LightGBM outperformed the other models in accuracy, especially when using simple statistical features such as the mean and standard deviation of EEG channels. This study highlights the integration of EEG signals with machine learning for stress monitoring, offering potential applications in wearable technology and brain-computer interfaces (BCIs).	Demonstrates high classification accuracy with LightGBM; emphasizes simple statistical features for stress detection; scalable for real-time applications in wearables and BCIs.	Limited generalizability as the study focuses on experimental stress induction rather than real-world scenarios; does not address potential issues of EEG noise or artifact interference.
Hafeez & Shakil (2024) ⁴⁶	This research examines stress in students during timed and untimed tests, correlating stress with EEG frequency bands such as beta, alpha, and theta. The study classified stress levels into three categories: relaxed, low, and high. Using deep learning frameworks, the study achieved 70.67% accuracy with LSTM and 90.46% accuracy with CNN. The findings suggest that time constraints increase stress and impair student performance. Furthermore, EEG frequency bands converted into brainwave images serve as effective biomarkers for stress detection.	Highlights EEG frequency bands as biomarkers for stress; achieves high accuracy with CNN; provides insights into the impact of time constraints on academic performance.	The study's artificial stress induction may not fully reflect natural stress responses; moderate accuracy with LSTM indicates room for methodological improvement.
Zhu et al. (2024) ⁴⁹	This study introduces a deep learning-based online assessment model for evaluating college students' mental health using textual comments from an online platform. A novel framework, CVCNN-MHFA, was developed and tested on 23,325 preprocessed comments. The model achieved a high accuracy of 94.56% with superior precision, recall, and F1 scores. Statistical analysis and visualization techniques confirmed the model's efficacy. The research establishes the model as a reliable tool for early detection of mental health issues, demonstrating its scalability and potential for institutional use.	Use textual data for mental health analysis; achieve excellent model performance metrics; demonstrates scalability and applicability for real-time monitoring.	Limited to textual data, potentially excluding other stress indicators like physiological signals; performance is dependent on the quality of online comments and preprocessing methods.
Kamakshamma & Bharati (2023) ⁵⁶	This study proposes an Adaptive-CSSA algorithm combined with Deep Belief Networks (DBNs) for predicting student dropout and stress levels. The MapReduce framework was used for efficient computation, and a feature selection strategy ensured accuracy. The model showed significant improvements in minimizing errors for both dropout and stress prediction. The results demonstrated the model's effectiveness in accurately predicting stress levels and identifying unique features from student performance data.	Employs an innovative feature selection approach; effectively reduces errors in predictions; integrates with MapReduce for computational efficiency.	Relies on a specific dataset, which may limit generalizability; high computational complexity might pose challenges for large-scale implementation.
Oryngozha et al. (2024) ⁵⁰	This study focuses on detecting stress-related posts in academic Reddit communities using natural language processing (NLP) and machine learning. The model was trained on the Dreddit dataset and achieved a 77.78% accuracy rate with Logistic Regression. A supplementary experiment validated the model with 72% accuracy on manually annotated academic subreddit posts. The study highlights that stress levels are higher in posts from professors' communities compared to students. This work provides insights into the psychological well-being of academic communities and emphasizes the importance of early intervention.	Uses social media data to capture real-world stress indicators; provides a novel focus on academic Reddit communities; highlights the psychological impact of academic environments.	Moderate accuracy limits the robustness of the model; reliance on text-based data may exclude other crucial indicators of stress.
Pourmohammadi & Maleki (2020) ⁴⁷	This comprehensive study examines stress detection using ECG and EMG signals from students under laboratory-induced stress conditions, such as time pressure and mental arithmetic. Data from 34 participants were analyzed, and stress recognition accuracies were reported as 100%, 97.6%, and 96.2% for two, three, and four stress levels, respectively. The study highlights the strong potential of EMG signals in combination with ECG for multi-level stress detection, demonstrating the effectiveness of physiological signal-based methods.	Demonstrates the effectiveness of physiological signals (ECG and EMG) for stress detection; achieves high accuracy in multi-level stress classification.	The experimental setup limits applicability to real-world scenarios; small sample size reduces the generalizability of the findings.
Figuerola et al. (2023) ⁵³	This research evaluates a multicomponent program aimed at managing academic stress through psychoeducation, mindfulness, and biofeedback. The study involved 38 high-achieving undergraduate students and assessed stress profiles using psychophysiological variables. Results showed a significant improvement in the Resilience to Stress Index (RSI) among participants, with 66% demonstrating better stress management after the intervention. The program was structured into three phases: pre-test evaluation, training, and post-test evaluation.	Provides evidence for the effectiveness of a multicomponent program; uses psychophysiological measures for comprehensive stress assessment; focuses on high-achieving students.	Limited sample size and demographic diversity; findings may not generalize to broader student populations.
Rois et al. (2021) ⁵⁷	This study used advanced machine learning approaches to predict stress prevalence among Bangladeshi university students. Data from 355 students were analyzed using algorithms like Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM). RF demonstrated the highest performance with an accuracy of 89.72%. Key stress predictors included pulse rate, blood pressure, sleep quality, and academic background. The findings suggest the value of ML models in identifying stress risk factors and guiding policy interventions in university settings.	Identifies significant predictors of stress; RF model achieves high accuracy; provides actionable insights for policy-making and mental health promotion.	A relatively small sample size may affect the reliability of results; limited geographical focus on Bangladeshi students restricts global applicability.
Ratul et al. (2023) ³⁷	This study investigates psychological and social stress levels in university students during the COVID-19 pandemic using machine learning. An online survey collected data from 444 ethnically diverse students, identifying high social stress in 11.26% of respondents and extremely high psychological stress in 24.10%. Machine learning models, including Multilayer Perceptron (MLP), achieved an accuracy of 80.5% with PCA for feature reduction and genetic algorithms for hyperparameter optimization. This work emphasizes the role of ML in stress detection and offers insights for mitigating stress during pandemics.	Uses real-world data from a diverse student sample; employs advanced techniques like PCA and genetic algorithms; highlights critical stress predictors.	Relies on self-reported data, which may introduce bias; convenience sampling limits generalizability; lacks long-term tracking of stress impacts.
Tseng et al. (2024) ⁵⁸	This longitudinal study uses EEG signals to detect stress in college students over 13 weeks. By employing Sequential Backward Selection (SBS) and Adaptive Synthetic (ADASYN) sampling, delta and theta EEG features were found to be highly indicative of stress. The leave-one-out cross-validation method demonstrated a balanced accuracy of 94.8%. ADASYN and Borderline-SMOTE methods improved the model's performance compared to traditional sampling methods. The study provides actionable insights into real-world stress monitoring and management.	Highlights EEG-based stress detection in real-world classroom settings; achieves high accuracy using advanced sampling and feature selection methods.	A limited sample size (18 participants) may affect generalizability; focuses solely on EEG signals, excluding other stress indicators.
Hantono et al. (2020) ⁵⁹	This research explores mental stress detection via heart rate variability (HRV) using smartphones during n-back tasks. HRV data was analyzed with machine learning algorithms, including k-Nearest Neighbor (kNN), which performed best in classifying time and frequency domain data. The study validates the use of smartphone sensors for stress detection and highlights the feasibility of using mobile devices for monitoring psychophysiological states in real-time.	Utilizes accessible and affordable smartphone technology for HRV measurement; demonstrates kNN as a reliable algorithm for stress classification.	Focuses on a small sample size (41 students); limited scope of tasks (n-back) may not generalize to diverse stress scenarios.
Continued			

Reference	Summary	Advantages	Disadvantages
Xu et al. (2024) ⁶⁰	This study introduces GOBO-DSNN, a hybrid deep spiking neural network optimized for student stress prediction. The model combines stochastic gradient descent and one-to-one optimization, simulating biological neural systems for improved accuracy. Evaluation metrics show high performance: accuracy of 90.976%, sensitivity of 91.698%, specificity of 91.336%, and precision of 90.179%. The framework addresses limitations of traditional models, offering a robust solution for early stress detection in students.	High performance metrics demonstrate reliability; integrates bio-inspired neural networks for enhanced accuracy; robust preprocessing methods for data cleaning.	Complexity of the model may limit scalability; reliance on specific datasets reduces generalizability; computationally intensive, potentially unsuitable for real-time applications.
AlShorman et al. (2022) ⁶¹	This study applies real-time EEG analysis of the frontal lobe for mental stress detection using machine learning. Features were extracted via Fast Fourier Transform (FFT), focusing on power density across EEG bands. Classification was performed using Support Vector Machine (SVM) and Naïve Bayes, achieving an average subject-wise accuracy of 98.21%. The low-complexity method shows promise for continuous stress monitoring and medical applications.	Achieves high accuracy with low computational complexity; demonstrates potential for real-time and continuous monitoring; easy to use and scalable.	Focuses only on the frontal lobe, potentially missing other relevant brain regions; does not address EEG signal artifacts or noise.
Liu & Wang (2022) ⁶²	This research focuses on detecting educational mental health stress using deep learning and computational intelligence optimization. A dataset specific to education was used to build a mental health stress detection model with emotional analysis functionality. Results showed strong model performance, particularly with the ant colony optimization (ACO) algorithm, which outperformed others in stability and execution time. Although results were promising, the study identifies areas for further optimization, particularly in addressing emotional states such as sadness and fear.	Employs advanced optimization techniques like ACO; emphasizes emotional analysis alongside stress detection; shows superior performance metrics compared to other algorithms.	Limited focus on specific emotional states, which reduces comprehensiveness; requires further optimization to address accuracy differences in detecting specific emotions like sadness and fear.
Fonda et al. (2024) ⁶³	This study develops a stacking ensemble model for stress level classification in higher education. Using physiological data (e.g., heart rate, body temperature), the model categorizes stress into anxious, calm, tense, and relaxed states. The stacking ensemble, enhanced with AdaBoost and XGBoost, achieved a 95% accuracy and an AUC of 0.95. The model excels in distinguishing extreme stress conditions and offers a reliable framework for improving stress detection accuracy and stability.	Achieves high accuracy and robust validation; effectively distinguishes between nuanced stress states; provides a scalable approach for higher education stress monitoring.	Focused on a specific institutional context (Hang Tuah University), which may limit generalizability; does not explore deep learning-based meta-models despite their potential.
Rajendran et al. (2024) ⁶⁴	This EEG-based study focuses on early stress detection in students using wavelet packet decomposition for feature extraction. EEG signals from 25 subjects were analyzed in relaxed and stressed states. The cubic Support Vector Machine (SVM) classifier outperformed others with an accuracy of 95.83%, sensitivity of 96.70%, specificity of 93.10%, and precision of 97.78%. The study highlights the efficacy of EEG band ratios and relative sub-band energy in stress classification, demonstrating the potential of machine learning for accurate stress detection.	Demonstrates high classification accuracy; validates EEG features for stress detection; provides actionable insights for early stress intervention.	Small sample size limits statistical significance; requires validation on larger, more diverse datasets; focused solely on EEG signals without considering multimodal data.
Pérez et al. (2018) ⁵¹	This study evaluates commercial off-the-shelf (COTS) wrist wearables for stress estimation in students. The research utilized a two-phase protocol involving controlled laboratory experiments and real-world classroom settings. Physiological data such as heart rate and skin temperature were collected, and stress levels were analyzed using a combination of quantitative and qualitative methods. Results suggest that COTS devices provide a feasible solution for monitoring stress in educational environments, though further validation is needed.	Demonstrates the feasibility of COTS devices for stress monitoring; combines laboratory and classroom settings for comprehensive evaluation; simple protocol for data collection.	Limited validation on a broader population; accuracy and reliability of COTS devices are not comparable to clinical-grade equipment; does not integrate advanced machine learning methods.
Luo et al. (2024) ⁵²	This study introduces Branched CALM-Net, a deep learning platform for predicting stress levels in college students using mobile sensor data. It addresses challenges like data irregularity, inter-subject variability, and the cold-start problem through dynamic clustering and multitask learning. Using data from the Dartmouth StudentLife study, the model achieved a 37% improvement in ROC AUC and surpassed baseline models in PR AUC. In cold start learning, it attained an F1 score of 87% with just one week of training data, demonstrating reliability and adaptability.	Addresses challenges like data irregularity and variability; achieves high accuracy and adaptability; incorporates personalization in stress prediction; demonstrates success with minimal training data.	Relies heavily on sensor data, limiting applicability in settings without such infrastructure; model complexity may hinder scalability or real-time deployment.
de Filippis & Foysal (2024) ⁵⁵	This study explores the primary stressors affecting students through machine learning. Using data from 1,100 students, psychological (self-esteem) and physiological (sleep quality) factors were identified as critical predictors of stress. A significant negative correlation was found between anxiety levels and academic performance. Random Forest models were employed for feature importance analysis, highlighting the relevance of targeted mental health interventions to improve student well-being and academic outcomes.	Identifies key stress factors through advanced machine learning techniques; emphasizes interventions targeting psychological and physiological stressors; uses a large, diverse dataset for robust analysis.	Limited to correlation analysis without exploring causal relationships; focuses primarily on stress factors without addressing mitigation strategies in detail.
Liao et al. (2024) ⁶⁵	This research proposes an intelligent perception-driven mental stress assessment method for college students. Factors from SRQ and SCL-90, including sleep, exercise, social interaction, and environment, were analyzed. Machine learning methods were employed to build an assessment model, with the Shapley Additive Explanations (SHAP) framework explaining the training results. The model effectively assessed students' mental stress, providing a foundation for non-intrusive and real-time stress detection.	Employs innovative intelligent sensing techniques for stress detection; integrates SHAP for explainability; achieves real-time and non-intrusive stress assessment.	Relies on specific intelligent sensing data, limiting generalizability; does not address long-term validation or the scalability of the proposed model.
Wali et al. (2022) ⁶⁶	This study uses EEG signals to detect stress by analyzing theta, alpha, and beta frequency bands. Data from 15 students undergoing mathematical exercises in noisy environments was processed with wavelet packet transform for feature extraction. The Extreme Learning Machine classifier achieved 98.56% accuracy with 92.52% sensitivity and 95.88% specificity. The research highlights the efficacy of EEG signals for stress detection, particularly in challenging environments.	High accuracy and sensitivity; demonstrates EEG effectiveness in noisy conditions; employs wavelet packet transform for precise feature extraction.	Limited sample size (15 students) reduces generalizability; focuses solely on EEG signals, omitting other potential indicators of stress.
Chen & Lee (2023) ⁶⁷	This study evaluates stress levels in university students using physiological signals (PPG, ECG, and EEG) collected during Sudoku-solving tasks under varying conditions. Enhanced deep learning models, including LRCN and self-supervised CNN, achieved high accuracy rates of up to 98.78% under comforting conditions. The findings outline the potential of integrating physiological signal analysis with real-world tasks for stress detection and monitoring in educational contexts.	Demonstrates high accuracy with advanced deep learning models; uses diverse stress-inducing scenarios for robust validation; integrates multiple physiological signals.	Focuses on specific tasks (Sudoku), which may not generalize to all stress scenarios; excludes long-term monitoring or evaluation of interventions.
Continued			

Reference	Summary	Advantages	Disadvantages
Tiwari & Agarwal (2021) ⁴⁸	This study develops a hybrid ANN-based model for pervasive stress detection using Galvanic Skin Response (GSR) and ECG signals. Data from 34 participants was collected during five experimental tasks, with stress levels classified into relaxed, stressed, partially stressed, and happy states. The model achieved 99.4% accuracy, outperforming other classifiers. The study highlights the potential of integrating IoT and real-time data analytics for accurate and continuous stress monitoring.	High classification accuracy; effectively addresses class imbalance using SMOTE; demonstrates integration of IoT for real-time stress detection.	Limited to 34 participants, reducing generalizability; focuses only on GSR and ECG signals, omitting multimodal data integration.
Anwar & Zakir (2022) ⁶⁸	This research proposes a low-cost, portable real-time mental stress detection method using Photoplethysmography (PPG). Data from 27 participants was analyzed to extract 36-time, frequency, and non-linear features. Machine learning classifiers achieved up to 81% accuracy using genetic algorithms for feature selection. The study highlights the feasibility of using affordable PPG devices for regular stress monitoring in daily life, reducing the reliance on clinical-grade equipment.	Demonstrates feasibility of using low-cost devices; integrates genetic algorithms for feature selection; emphasizes real-life applicability.	Small sample size limits robustness; accuracy levels may not meet clinical standards; excludes validation on diverse populations.
Morales-Rodríguez et al. (2021) ⁵⁴	This study examines stress, coping strategies, and resilience among university students during and after the COVID-19 pandemic using AI-based predictive models. Data from 337 participants revealed associations between resilience, coping strategies, and stress levels. The neural network model achieved an 80% predictive capacity. Findings emphasize the importance of resilience in managing stress and provide insights for designing specific interventions to support students' mental health during crises.	Highlights resilience as a key factor in stress management; integrates AI for predictive modeling; offers actionable insights for educational psychology.	Limited to a specific context (COVID-19 pandemic); small sample size reduces the generalizability of findings; focuses primarily on predictive capacity without exploring implementation strategies for interventions.

Table 1. Comparative analysis of stress detection studies in students: summary, advantages, and disadvantages.

Material and method

The foundation of this study lies in its comprehensive approach to data collection and analysis, aimed at uncovering the multifaceted factors contributing to student stress. The dataset assembled encompasses a diverse array of variables, including demographic details, academic performance metrics, and social factors, each potentially playing a role in influencing student stress levels. Prior to analysis, this data was preprocessed to ensure accuracy and relevance. Since, the data is taken from a previously existing database⁶⁹, and the focus of this article is on the application of artificial intelligence methods^{70,71}, consequently, the experimental protocol is not the scope of the article.

The section follows a structured supervised machine learning pipeline for educational data mining, as illustrated in Fig. 1, to analyze and interpret student stress data. The process begins with a curated dataset of 1,100 student responses, encompassing 21 features across psychological, physiological, social, environmental, and academic domains. The pipeline includes data normalization and multicollinearity diagnostics using the Variance Inflation Factor (VIF) to ensure statistical soundness. It applies six supervised classification algorithms—Logistic Regression, Support Vector Machine, Gradient Boosting, Decision Tree, XGBoost, and Random Forest—to model stress levels. It further uses grid search and validation curves for hyperparameter optimization, enhancing model accuracy and generalizability. Finally, SHAP analysis interprets the influence of each feature on the model outputs, supporting transparent, data-driven insights for educational innovation.

Dataset

The dataset utilized in this study⁶⁹ offers an insightful glimpse into the diverse factors contributing to student stress. Encompassing 21 features, it integrates elements from five key domains impacting student well-being: (1) psychological, (2) physiological, (3) social, (4) environmental, and (5) academic. This range spans mental health aspects, physiological markers, academic experiences, and the quality of social interactions and environmental conditions. The database has also been used by de Filippis & Foysal (2024)⁵⁵ quoting it to be “*rigorous survey conducted across various educational institutions, capturing a wide demographic spectrum. It comprises detailed responses from 1100 students, meticulously assembled to explore the multifaceted nature of stressors within academic environments*”.

Figure 2 presents a scatter matrix that visualizes pairwise relationships and distributions among all predictor variables, grouped by stress levels (0, 1, and 2). Each diagonal plot shows the distribution of individual features, while the off-diagonal scatter plots reveal potential linear or non-linear associations between variables. The matrix highlights visible clustering patterns and separation trends for certain features across stress levels, supporting their relevance for classification tasks. By examining these interdependencies, the scatter matrix aids in identifying correlated variables and assessing their discriminatory power in predicting student stress, thereby guiding both feature selection and model interpretability.

The bar chart in Fig. 3 explains the distribution of stress levels across a population sample, indicating a nearly uniform spread with a total of 373, 358, and 369 instances for stress levels 0, 1, and 2 respectively. This distribution provides an insight into the prevalence of various stress intensities within the cohort and suggests a balanced representation across the spectrum of stress categorization. The consistency in sample size for each stress level underscores the reliability of the dataset for further statistical analysis and could potentially reflect the effectiveness of the sampling methodology in capturing a diverse set of responses pertaining to stress experiences. The distribution of stress levels depicted in the bar chart (Fig. 3) is of particular importance for the application of machine learning methods, as it reflects a balanced dataset that is crucial for the development of unbiased and generalizable predictive models. Balanced datasets are paramount in machine learning to prevent overfitting and to ensure that the model performs well not only on the training data but also on unseen data. In this context, the similar number of samples for stress levels 0, 1, and 2 suggests that the dataset is well-prepared for algorithms that require uniform representation of classes, such as many supervised learning techniques.

Data-driven decision-making for educational innovation policies in higher education



SHAP Analysis

Feature importance rankings and SHAP values were computed to interpret and visualize the influence of each predictor on model outputs, allowing for transparent assessment of the variables contributing to student stress predictions.

Hyperparameter Optimization

Hyperparameters for all models were tuned using grid search and validation curves to identify configurations that maximized cross-validated classification accuracy.

Classification

Six supervised learning algorithms—Logistic Regression, Support Vector Machine, Gradient Boosting, Decision Tree, XGBoost, and Random Forest—were employed to model and predict student stress levels.

Pre-processing

The data underwent normalization and multicollinearity assessment using Variance Inflation Factor (VIF) to ensure statistical integrity and model interpretability.

Dataset

The study utilized a dataset comprising 1,100 student responses across 21 variables that span psychological, physiological, social, environmental, and academic domains.

Raw Data

Fig. 1. Supervised Machine Learning Pipeline for Educational Data Mining (EDM), illustrating the step-by-step methodology used to analyze student stress data—from raw dataset preparation and pre-processing, through classification and hyperparameter optimization, to SHAP-based model interpretation—supporting data-driven strategies for educational innovation in higher education.

Moreover, this balance across categories may contribute to more accurate cross-validation results during the model training phase, leading to robust machine learning models that can reliably predict stress levels, thereby providing valuable insights into the stress patterns within the targeted demographic. As can be observed in the bar chart in Fig. 3, the distribution of the dataset across the three stress levels: low (0), medium (1), and high (2), is very balanced. The nearly uniform distribution across these categories ensures a balanced representation, which is critical for the robustness of subsequent classification analyses and the generalizability of the results. This balance in data also outlines the prevalence of varied stress experiences among students, reinforcing the need for differentiated approaches in stress mitigation strategies.

On the other hand, Fig. 4 offers a comprehensive correlation matrix based on Pearson coefficients, which examines the pairwise relationships between all variables within the dataset. The heatmap representation of a correlation matrix also details the pairwise correlation coefficients between various factors related to stress. In the context of statistical analysis and machine learning, such a matrix is instrumental for identifying relationships between variables, which can be critical for feature selection and for understanding the underlying structure of the data. This matrix serves as a heatmap that not only highlights the strength and direction of associations between the factors but also reveals potential multicollinearity, which is instrumental in refining the models for accuracy. The color-coded visualization provides immediate insights into how closely connected certain stress indicators are, such as the link between 'study_load' and 'academic_performance' or 'sleep_quality' and 'mental_health_history', thereby laying the groundwork for a more targeted analysis in the feature selection phase of model building. The rows and columns of the matrix correspond to different variables, such as 'anxiety_level',

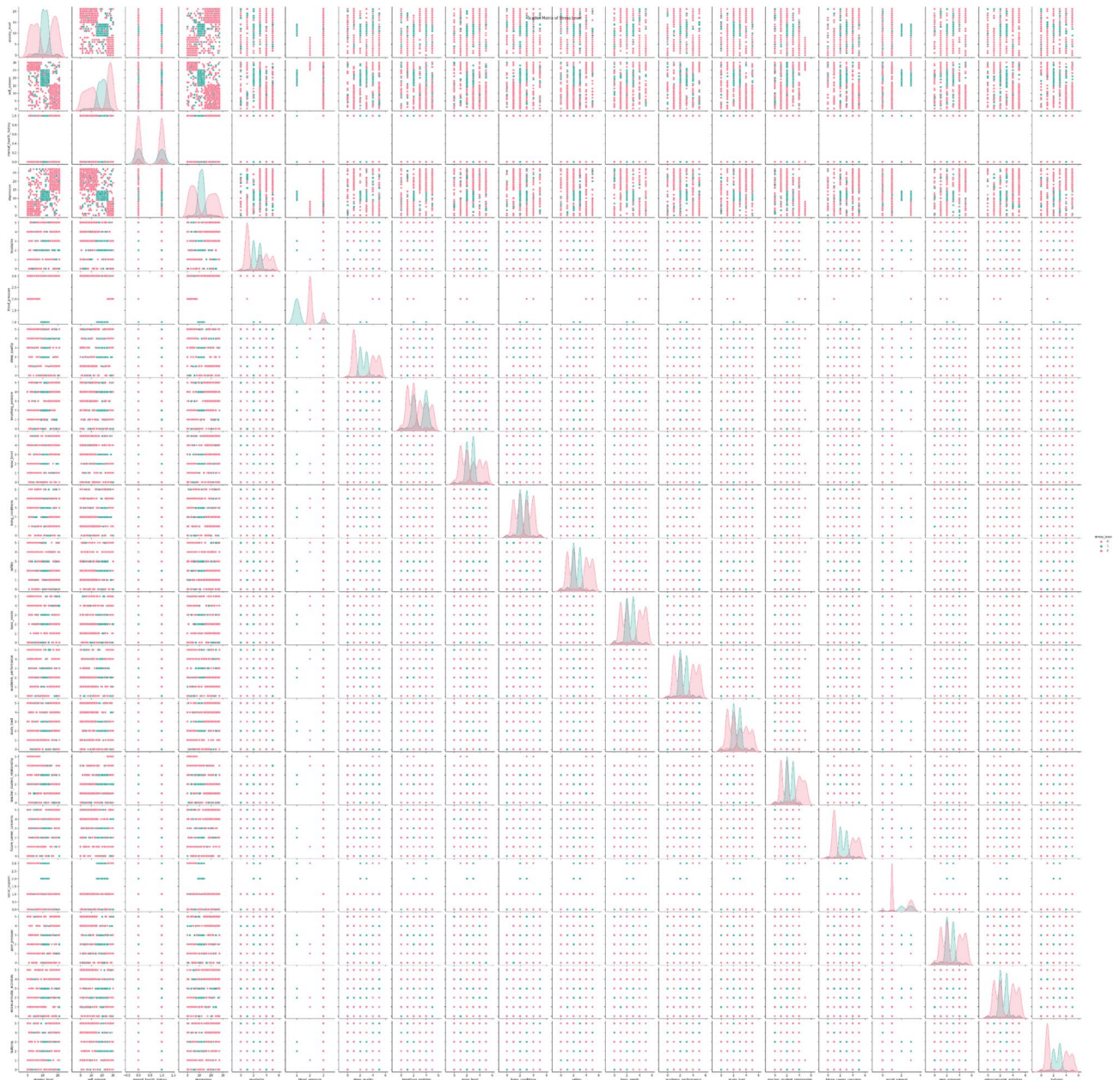


Fig. 2. Scatter matrix of the data variables. The full figure is available on the website: <https://doi.org/10.6084/m9.figshare.24632076>.

'self_esteem', 'mental_health_history', 'depression', and 'blood_pressure', among others. Each cell within the matrix represents the correlation coefficient between the variables on its corresponding row and column, with values ranging from -1 to $+1$. A value of $+1$ indicates a perfect positive correlation, meaning that as one variable increases, the other variable also increases. Conversely, a value of -1 indicates a perfect negative correlation, implying that as one variable increases, the other decreases. Values close to 0 suggest no linear correlation. Colors ranging from blue to red represent the strength and direction of the correlation, with blue indicating positive correlations and red indicating negative correlations. For instance, the variables 'anxiety_level' and 'stress_level' are shown to have a strong positive correlation, as indicated by a deep blue color and a coefficient close to 1 . In contrast, 'self_esteem' and 'stress_level' display a strong negative correlation, denoted by a deep red color and a coefficient near -1 . Self-esteem is inversely correlated with stress level, indicating that lower self-esteem may be associated with higher stress. Mental health history also shows a negative correlation, suggesting that a history of mental health issues might contribute to elevated stress levels. Depression is strongly and positively correlated with stress, as expected, with both likely to increase concurrently. Headaches, often a somatic symptom of stress, exhibit a positive correlation, reinforcing this common psychophysiological linkage. Blood pressure, which can be affected by chronic stress, shows a moderate positive correlation. Sleep quality is inversely correlated with stress, supporting the notion that poor sleep can be both a cause and a consequence of stress. Breathing

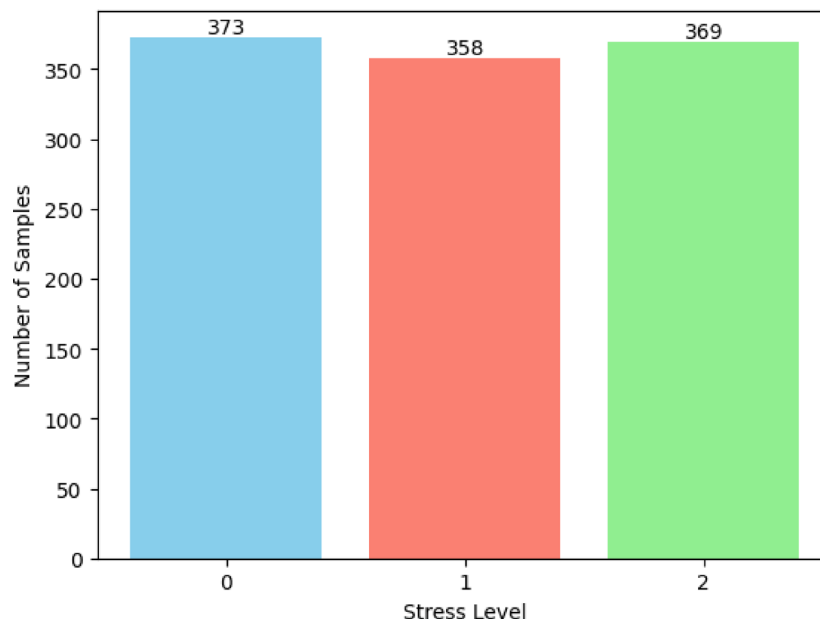


Fig. 3. The number of samples in the database which correspond to the stress level: 0, 1, and 2.

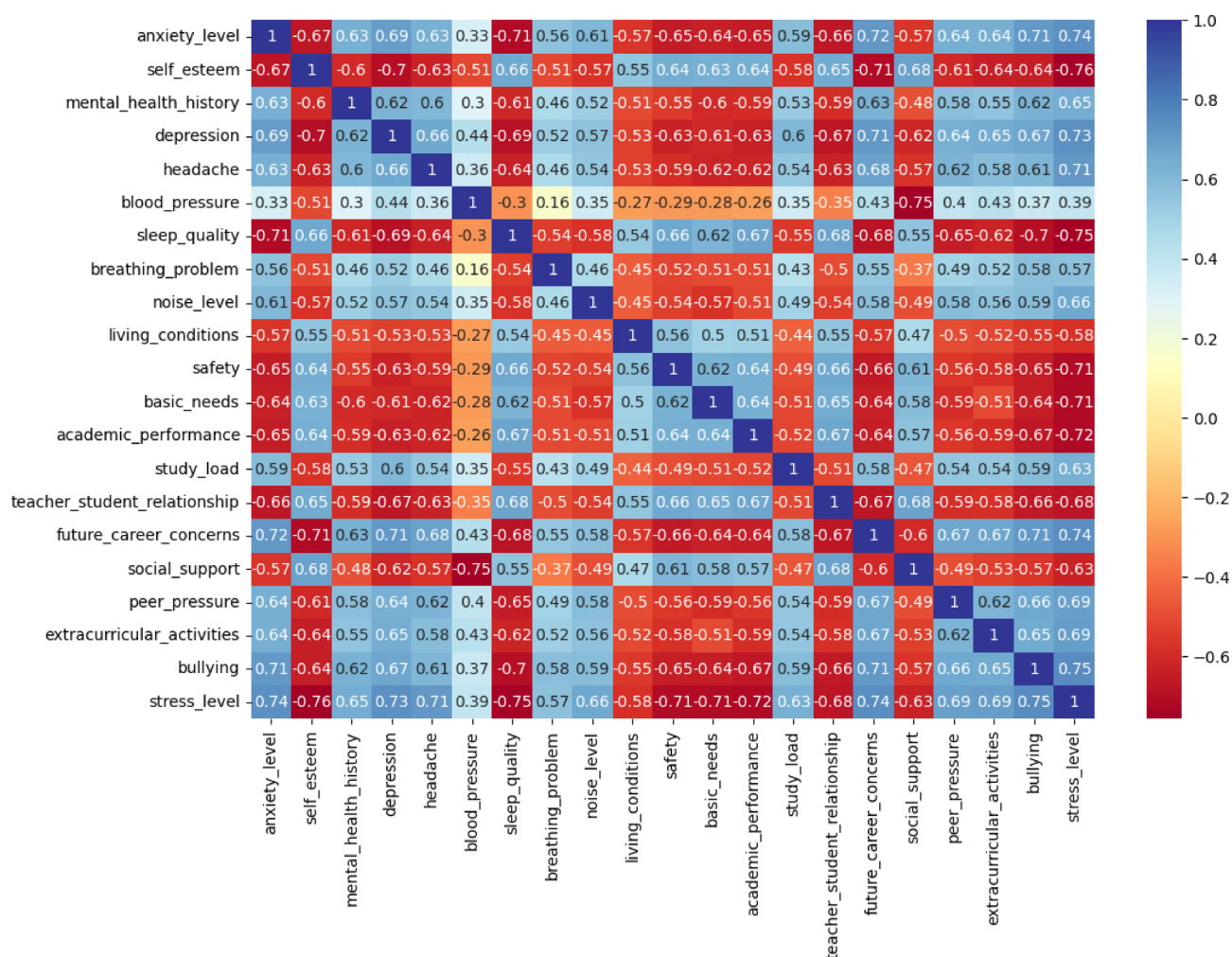


Fig. 4. Correlation matrix based upon Pearson between all the variables of the database.

problems show a positive correlation with stress, possibly reflecting the physiological effects of anxiety and stress on respiratory patterns. Noise level has a negative correlation, suggesting that higher noise levels might be linked to increased stress. Living conditions and safety both display negative correlations with stress, highlighting the impact of one's environment on their stress levels. The fulfillment of basic needs is inversely related to stress, which aligns with the understanding that unmet basic needs can be a significant source of stress. Academic performance shows a negative correlation, indicating that lower performance might correlate with higher stress levels among students. Study load is positively correlated with stress, a finding that is intuitive as a heavier study load can increase stress. The teacher-student relationship is negatively correlated, which could imply that positive relationships with teachers might buffer or reduce students' stress levels. Future career concerns have a strong positive correlation with stress, reflecting the anxiety that uncertainty regarding future employment can induce. Social support is shown to have a negative correlation, emphasizing its role as a protective factor against stress. Peer pressure is positively correlated, suggesting that negative interactions with peers could exacerbate stress. Extracurricular activities present a negative correlation, possibly indicating that engagement in such activities may reduce stress or provide a constructive outlet for stress relief. Lastly, bullying has a strong positive correlation with stress level, underscoring the severe impact of bullying on an individual's stress and overall well-being. Each of these correlations provides insight into potential areas of intervention and the multifactorial nature of stress, which is critical for designing targeted strategies to mitigate stress and its adverse effects on individuals.

Pre-processing

Pre-processing is a critical step in ensuring the reliability and interpretability of machine learning models, especially when dealing with high-dimensional educational and psychological datasets⁷². In predictive modeling, the presence of multicollinearity among predictor variables can distort the significance of individual predictors, inflate standard errors, and reduce model generalizability⁷³. To mitigate such statistical distortion and enhance the interpretability of regression-based or classifiers, multicollinearity diagnostics were conducted before model training. This step aimed to preserve the mathematical integrity of the predictors and ensure that each feature contributed uniquely to the model's classification of stress levels.

To assess multicollinearity, the Variance Inflation Factor (VIF) was calculated for each predictor variable, excluding the target variable *stress_level*. The implementation followed a standard statistical procedure using the *statsmodels* package in Python. The independent variables were first isolated, and a constant was added to the feature matrix using *add_constant()* to account for the intercept term. The VIF for each variable was then computed with *variance_inflation_factor(X.values, i)* over all columns in the feature matrix X. The resulting VIF values quantify how much the variance of an estimated regression coefficient increases if predictors are correlated. Values near 1 indicate no multicollinearity, while increasing values suggest stronger intercorrelation among features. However, it is generally agreed that $VIF < 5$ typically indicates low multicollinearity, whereas VIF between 5 and 10 indicates moderate multicollinearity, however, it can be acceptable depending on the specific application, research context, and whether variables are essential to the theoretical model, nevertheless, $VIF > 10$ indicates high multicollinearity; problematic and usually requires corrective action (e.g., removing variables, combining predictors, or using regularization)^{73,74}.

The results of this analysis are reported in Table 2, which lists the VIF values for all 21 predictor variables. The constant term exhibits a high VIF of 139.64, which is expected due to its linear dependency with the sum of all predictors and does not indicate a multicollinearity problem⁷⁵. All remaining variables exhibit VIF values below the critical threshold of 10. Notably, *social_support* presents the highest VIF among the predictors at 5.75, followed by *blood_pressure* at 3.69, *future_career_concerns* at 3.42, and *anxiety_level* at 3.23. These values suggest a moderate level of multicollinearity, but none indicate a severe issue that would necessitate variable exclusion. The majority of predictors fall in the range of 1.78 to 3.20, supporting their inclusion in the model as independent contributors to the prediction of stress levels.

Despite the moderate multicollinearity associated with *social_support*, it was retained as a predictor variable due to both its empirical relevance and theoretical justification. As shown in Table 2, its VIF value of 5.75 lies close to the lower bound of moderate multicollinearity (typically considered between 5 and 10), and does not exceed any critical threshold that warrants exclusion. According to Hair et al. (2010) in *Multivariate Data Analysis*⁷³, a VIF below 10 is generally acceptable, and values between 5 and 10 can be tolerated in cases where the variable is conceptually important and theoretically justified. Removing this variable would not only limit the model's explanatory power but also neglect a critical domain of psychological resilience in the context of student stress. Therefore, based on both statistical justification and theoretical validity, *social_support* was retained as an explanatory feature in the predictive modeling process.

Classification through machine learning methods

The classification of student stress levels was programming on Python environment using Google Colab and was approached using a structured supervised machine learning for educational data mining. The dataset was first split into training and testing subsets using an 80:20 ratio to ensure that model evaluation was conducted on unseen data. Multiple classification algorithms, including Logistic Regression, Support Vector Machine, Gradient Boosting, Decision Tree, XGBoost, and Random Forest were implemented.

The study employed six supervised machine learning algorithms to classify student stress levels, each offering distinct strengths in pattern recognition and model interpretability. Logistic Regression is a linear model that estimates the probability of categorical outcomes based on input features by fitting data to a logistic function⁷⁶. Support Vector Machine (SVM) constructs an optimal hyperplane to separate classes in a high-dimensional space, using kernel functions to handle non-linear relationships⁷⁷. Gradient Boosting constructs a strong model through a sequence of weak learners, where each learner focuses on minimizing the errors of its predecessor⁷⁸.

Feature	VIF
const	139.64
anxiety_level	3.23
self_esteem	3.21
mental_health_history	2.22
depression	3.09
headache	2.53
blood_pressure	3.69
sleep_quality	3.09
breathing_problem	1.78
noise_level	1.99
living_conditions	1.80
safety	2.79
basic_needs	2.68
academic_performance	2.71
study_load	1.90
teacher_student_relationship	3.20
future_career_concerns	3.42
social_support	5.75
peer_pressure	2.61
extracurricular_activities	2.52
bullying	3.19

Table 2. Variance inflation factor (VIF) values for predictor variables in the stress level model.

Decision Tree (DT) operates by recursively splitting the data based on feature values, creating an interpretable tree structure of decision rules⁸. XGBoost (Extreme Gradient Boosting) is an advanced form of gradient boosting optimized for speed and performance, incorporating regularization to prevent overfitting⁷⁹. Random Forest enhances the robustness of decision trees by building an ensemble of trees using bootstrapped samples and aggregating their predictions to reduce overfitting^{80,81}.

Each model underwent hyperparameter optimization through *GridSearchCV* using 5-fold cross-validation on the training set to identify the optimal configurations that maximize predictive accuracy. The performance of the trained models was evaluated using key metrics such as accuracy, precision, recall, and F1-score, derived from the classification report, alongside the confusion matrix to visualize prediction outcomes across the three stress categories. Additionally, validation curves were generated to examine the effect of varying individual hyperparameters on model performance, offering insights into the stability and generalization of each classifier.

Grid search and hyperparameter optimization through validation curves

Hyperparameters are external configurations of machine learning models that govern their learning process and structural behavior but are not learned from the data itself⁸². Identifying optimal hyperparameters is essential because they directly influence model performance, generalization, and the trade-off between underfitting and overfitting⁸³. In this study, grid search was employed to systematically explore combinations of hyperparameters for each classifier using cross-validated accuracy as the evaluation metric. Table 3 presents the grid search spaces alongside the best-performing hyperparameter configurations for six machine learning algorithms: Logistic Regression, Support Vector Machine (SVM), Gradient Boosting, Decision Tree, XGBoost, and Random Forest. For instance, Logistic Regression achieved optimal results with *penalty* = 'l2' and *C* = 10, while the SVM model performed best with a linear kernel and *C* = 10. Gradient Boosting benefited from a low *learning_rate* = 0.01 and *max_depth* = 7, whereas the Decision Tree classifier was most effective with *max_depth* = 10 and *min_samples_split* = 20. The XGBoost model was optimized with *n_estimators* = 150, *max_depth* = 6, and *learning_rate* = 0.05, and the Random Forest model yielded the best results using *n_estimators* = 150 and *max_depth* = 10. These configurations were selected based on their ability to maximize predictive accuracy and were validated using both performance metrics and visualization of validation curves.

Figure 5 shows the validation curves used to tune hyperparameters for Logistic Regression, confirming the selection of *penalty* = 'l2' and *C* = 10 as the best configuration listed in Table 3. In Fig. 5a, the l2 penalty achieves the highest mean cross-validated accuracy compared to l1 and *elasticnet*, indicating its superiority for regularization in this context. Figure 5b demonstrates that increasing the inverse regularization strength (*C*) enhances accuracy up to a stable peak at *C* = 10, supporting its selection. Figure 5c shows that variation in *l1_ratio* within the *elasticnet* penalty results in slightly lower accuracy, further justifying the exclusion of *elasticnet* in favor of l2. These curves validate the grid search results and reinforce the robustness of the selected hyperparameters for Logistic Regression in predicting student stress levels.

Figure 6 presents the validation curves used to optimize the hyperparameters for the Support Vector Machine model, validating the best configuration shown in Table 3. In Fig. 6a, accuracy improves with increasing regularization strength (*C*), stabilizing at *C* = 10, which confirms its selection. Figure 6b indicates that the linear

Algorithm	Grid search space	Best hyperparameters
Logistic regression	<i>penalty</i> = ['l1', 'l2', 'elasticnet', 'none'] <i>C</i> = [0.001, 0.01, 0.1, 1, 10, 100] <i>l1_ratio</i> = [0, 0.25, 0.5, 0.75, 1]	<i>penalty</i> = 'l2' <i>C</i> = 10
Support vector machine	<i>C</i> = [0.01, 0.1, 1, 10, 100] <i>kernel</i> = ['linear', 'rbf', 'poly', 'sigmoid'] <i>gamma</i> = ['scale', 'auto'] <i>degree</i> = [2, 3, 4]	<i>C</i> = 10 <i>kernel</i> = 'linear' <i>gamma</i> = 'scale'
Gradient boosting	<i>n_estimators</i> = [50, 100, 150] <i>learning_rate</i> = [0.01, 0.05, 0.1, 0.2] <i>max_depth</i> = [3, 5, 7] <i>min_samples_split</i> = [2, 5, 10] <i>min_samples_leaf</i> = [1, 2, 4] <i>subsample</i> = [0.6, 0.8, 1.0]	<i>n_estimators</i> = 50 <i>learning_rate</i> = 0.01 <i>max_depth</i> = 7 <i>min_samples_split</i> = 2 <i>min_samples_leaf</i> = 1 <i>subsample</i> = 1.0
Decision tree	<i>max_depth</i> = [3, 5, 10, 15, 20, 25, None] <i>min_samples_split</i> = [2, 5, 10, 15, 20] <i>min_samples_leaf</i> = [1, 2, 4, 6, 8] <i>max_features</i> = [None, 'sqrt', 'log2'] <i>max_leaf_nodes</i> = [None, 10, 20, 30, 40] <i>criterion</i> = ['gini', 'entropy']	<i>max_depth</i> = 10 <i>min_samples_split</i> = 20 <i>min_samples_leaf</i> = 6 <i>max_features</i> = 'sqrt' <i>max_leaf_nodes</i> = None <i>criterion</i> = 'gini'
XGBoost	<i>n_estimators</i> = [150, 200, 250] <i>max_depth</i> = [6, 8, 10] <i>learning_rate</i> = [0.01, 0.05, 0.1] <i>subsample</i> = [0.8, 1.0] <i>colsample_bytree</i> = [0.8, 1.0]	<i>n_estimators</i> = 150 <i>max_depth</i> = 6 <i>learning_rate</i> = 0.05 <i>subsample</i> = 0.8 <i>colsample_bytree</i> = 1.0
Random forest	<i>n_estimators</i> = [150, 200, 250] <i>max_depth</i> = [10, 12, 15, 18] <i>min_samples_split</i> = [2, 5, 10] <i>min_samples_leaf</i> = [1, 2, 3] <i>max_features</i> = ['sqrt', 'log2']	<i>n_estimators</i> = 150 <i>max_depth</i> = 10 <i>min_samples_split</i> = 10 <i>min_samples_leaf</i> = 1 <i>max_features</i> = None

Table 3. Summary of best hyperparameter configurations for each machine learning algorithm evaluated in the study: logistic regression, support vector machine, gradient Boosting, decision tree, XGBoost, and random forest. The hyperparameters were optimized to enhance classification accuracy in predicting student stress levels.

kernel outperforms poly, rbf, and sigmoid, with the sigmoid showing the poorest performance, supporting the choice of *kernel* = 'linear'. Figure 6c demonstrates that the scale setting for the kernel coefficient (*gamma*) yields higher accuracy than auto, justifying *gamma* = 'scale'. Finally, Fig. 6d shows that changes in polynomial degree produce no gain in accuracy, reinforcing that a non-polynomial kernel such as linear is more effective. These results support the grid search findings and confirm that *C* = 10, *kernel* = 'linear', and *gamma* = 'scale' offer optimal performance for stress level classification using SVM.

Figure 7 displays the validation curves used to optimize hyperparameters for the Gradient Boosting classifier, confirming the final configuration reported in Table 3. In Fig. 7a, the model achieves peak accuracy with 50 estimators, indicating that additional trees provide no added benefit. Figure 7b shows that a lower learning rate of 0.01 results in the highest stability and performance, reinforcing its selection for gradual, accurate learning. Figure 7c reveals that *maximum depth* = 7 produces a slight edge over shallower configurations, providing adequate model complexity. Figure 7d demonstrates optimal performance at *min_samples_split* = 2, while Fig. 7e confirms that *min_samples_leaf* = 1 supports the best accuracy, allowing fine-grained splitting. Lastly, Fig. 7f indicates that using a subsample ratio of 1.0 (i.e., full dataset per iteration) achieves the best balance between variance reduction and bias control. Together, these curves validate the hyperparameter set used to train the Gradient Boosting model for student stress classification.

Figure 8 presents the validation curves for the Decision Tree classifier, which support the hyperparameter configuration summarized in Table 3. In Fig. 8a, accuracy peaks at a maximum tree depth of 10, balancing complexity and generalization. Figure 8b shows that the best performance occurs when the minimum samples required to split is 20, helping to prevent overfitting. Figure 8c indicates that setting minimum samples per leaf to 6 yields better generalization by avoiding overly specific branches. Figure 8d demonstrates that selecting 'sqrt' for the maximum number of features at each split leads to slightly improved accuracy over log2. In Fig. 8e, the maximum number of leaf nodes set to None performs comparably or better than fixed values, enabling unrestricted tree growth when necessary. Lastly, Fig. 8f confirms that *gini* is the preferred splitting criterion, outperforming entropy. Together, these curves validate the optimized Decision Tree settings used to classify stress levels accurately.

Figure 9 illustrates the validation curves for the XGBoost classifier, confirming the optimal hyperparameter configuration reported in Table 3. In Fig. 9a, the classifier performs best with 150 estimators, beyond which accuracy plateaus. Figure 9b shows that a maximum depth of 6 yields the highest accuracy, indicating sufficient model complexity without overfitting. Figure 9c confirms that a learning rate of 0.05 strikes the best balance between convergence speed and stability. Figure 9d demonstrates that a subsample ratio of 0.8 provides slightly better performance than using the full dataset, supporting enhanced generalization. Lastly, Fig. 9e reveals that setting *colsample_bytree* to 1.0 (using all features per tree) leads to the highest cross-validated accuracy. These curves collectively validate the selected hyperparameters for training the XGBoost model to classify student stress levels with high precision.

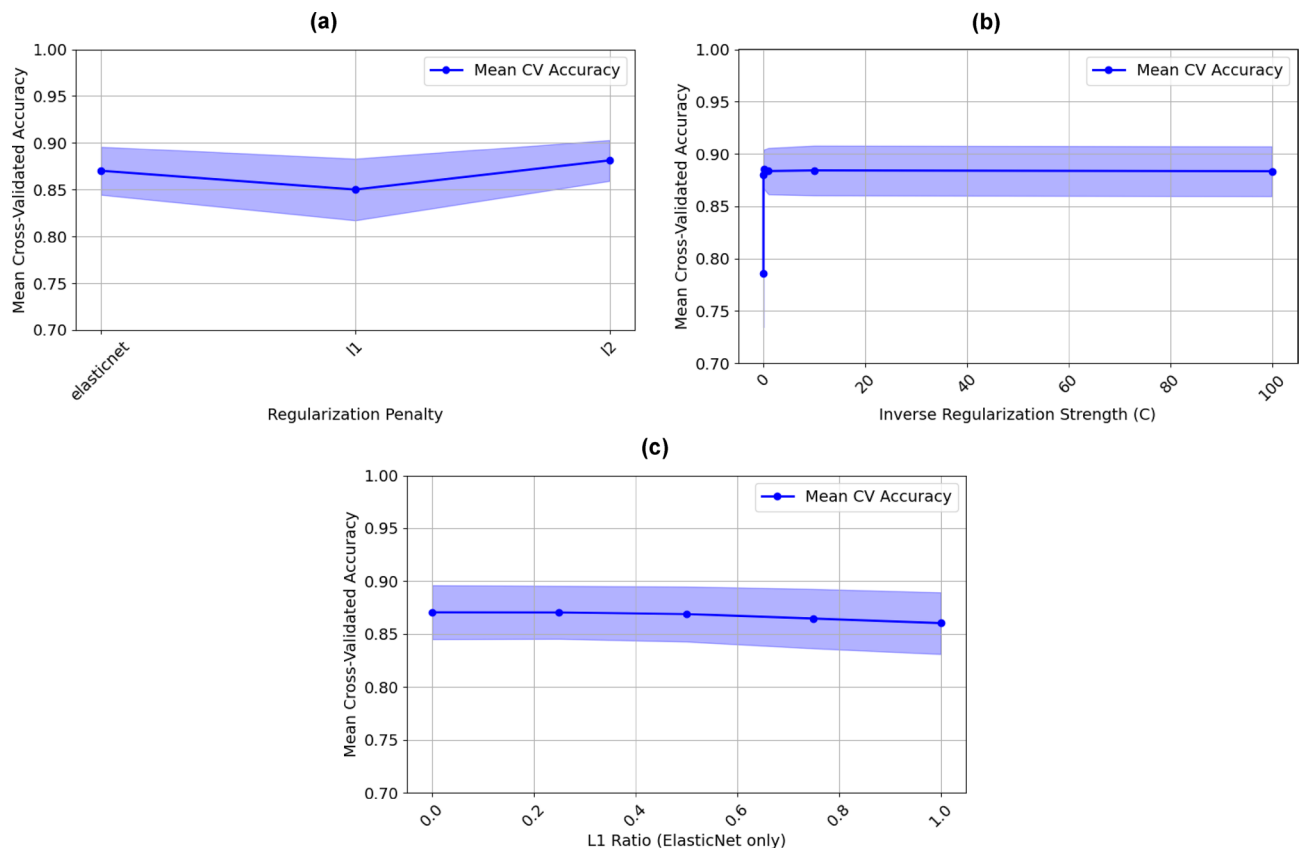


Fig. 5. Validation curves for Logistic Regression showing the mean cross-validated accuracy across different regularization settings: (a) penalty types (*elasticnet*, *l1*, *l2*), (b) inverse regularization strength (*C*), and (c) L1 ratio values used in the *elasticnet* penalty.

Figure 10 shows the validation curves for the Random Forest classifier and supports the hyperparameter configuration presented in Table 3. In Fig. 10a, the classifier achieves optimal accuracy with 150 estimators, with no significant gains beyond that. Figure 10b confirms that a maximum depth of 10 provides the best performance by balancing complexity and generalization. Figure 10c identifies *min_samples_split* = 10 as the most stable and accurate setting, helping to control overfitting. In Fig. 10d, the highest accuracy is reached when the minimum number of samples per leaf is 1, allowing for detailed decision boundaries. Finally, Fig. 10e shows that no restriction on *max_features* (i.e., using all features) slightly outperforms *sqrt* or *log2*, confirming that considering all features per split improves model accuracy. These results validate the final Random Forest configuration for classifying student stress levels.

Feature importance analysis and Shapley Additive exPlanations (SHAP)

XAI aims to make machine learning models transparent by clarifying how inputs influence predictions^{8,84,85}. SHAP (SHapley Additive exPlanations) is an XAI method because it provides a unified, model-agnostic framework to interpret predictions by attributing each feature's contribution in a transparent and mathematically consistent way⁸⁶. It works by computing the Shapley value for each feature, a concept from cooperative game theory, which fairly distributes the “payout” (in this case, the prediction) among all features by considering all possible combinations in which the feature could be added to the model. SHAP calculates the average marginal contribution of each feature across all permutations, ensuring consistency and local accuracy. This method allows researchers to understand which features matter most and how they influence individual predictions—whether they push the prediction higher or lower. It is particularly suitable for tree-based models⁸⁷ such as Decision Tree, Random Forest, XGBoost, and Gradient Boosting because these models allow fast, exact SHAP value computation using *TreeExplainer*. This makes SHAP a powerful tool in educational data mining where transparency and interpretability are crucial for responsible decision-making.

Results

Performance indicators, including precision, recall, F1-score, and accuracy, were chosen to evaluate the models' effectiveness. Precision, denoting the accuracy of positive predictions, is calculated as the ratio of true positive predictions to the sum of true positive and false positive predictions. Recall assesses the model's ability to identify every positive event, calculated as the ratio of true positive predictions to the sum of true positive and false negative predictions. The F1-score, a harmonic means of precision and recall, strikes a balance between these

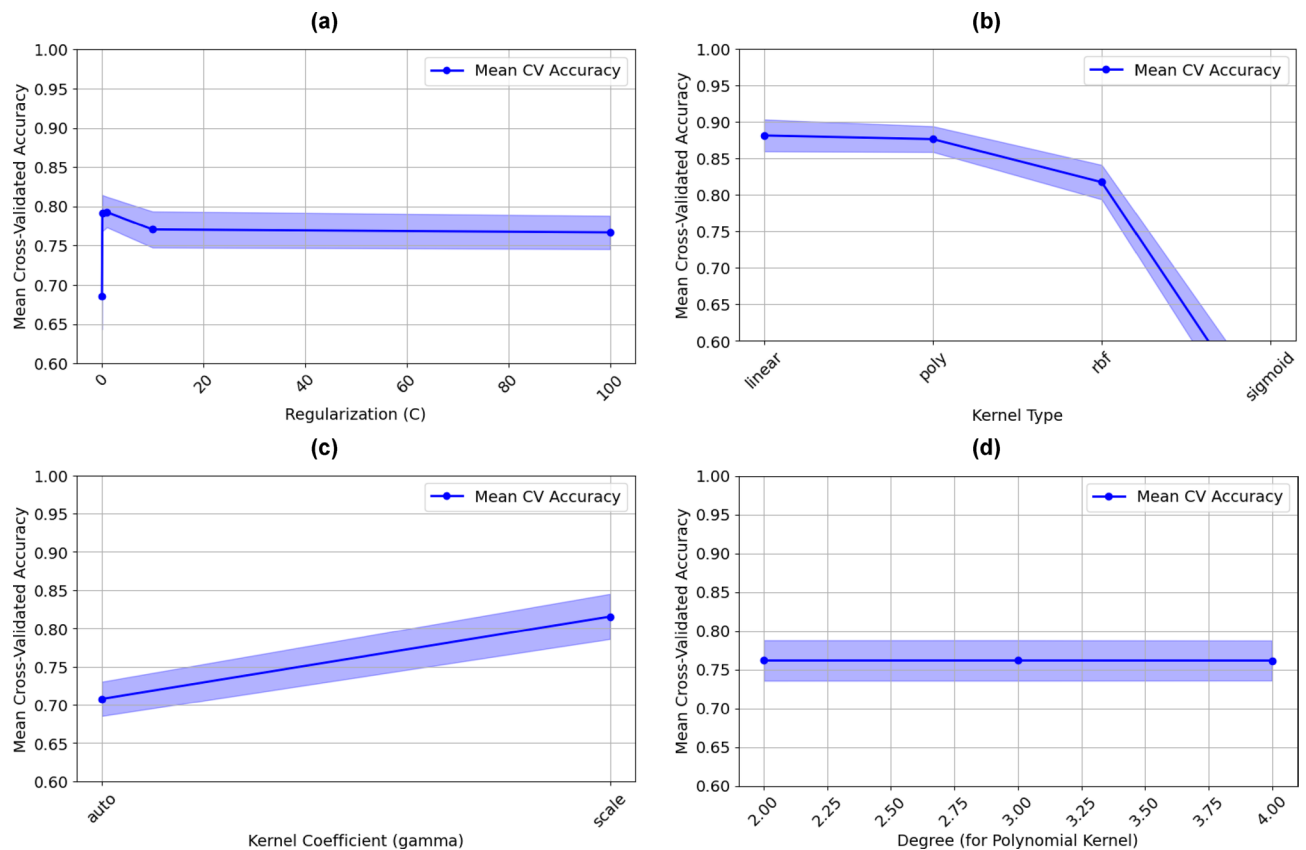


Fig. 6. Validation curves for Support Vector Machine showing the mean cross-validated accuracy across key hyperparameters: **(a)** regularization strength (C), **(b)** kernel type, **(c)** kernel coefficient (gamma), and **(d)** polynomial degree for the polynomial kernel.

two metrics, offering a comprehensive measure of a model's performance. Finally, accuracy measures the overall correctness of predictions, calculated as the ratio of the sum of true positive and true negative predictions to the total number of predictions. In conclusion, the results of these machine learning algorithms for the specified performance indicators are comprehensively presented in the following Table 4.

Table 4 presents the classification reports for six machine learning models, comparing their performance in predicting three levels of student stress. Subtable (a) shows that Logistic Regression achieves an overall accuracy of 0.89, with particularly strong results for Class 2 (F1-score = 0.92), indicating its balanced performance across classes. In (b), Support Vector Machine matches this overall accuracy of 0.89 and exhibits consistent metrics across all classes, confirming its robustness. Subtable (c) shows that Gradient Boosting yields a slightly lower accuracy of 0.87, with balanced but slightly weaker performance across classes, particularly for Class 2. Subtable (d), representing Decision Tree, also achieves 0.87 accuracy, showing marginally lower recall for Class 2, suggesting slightly less consistency in classification. In (e), XGBoost improves the performance with an accuracy of 0.88 and uniformly high precision and recall across all classes, reflecting its strong generalization. Finally, subtable (f) shows that Random Forest also achieves 0.88 accuracy, with particularly high precision for Class 1 (0.93), making it effective for detecting mid-level stress cases. Collectively, these results confirm that Logistic Regression and SVM slightly outperform other models, while ensemble methods like XGBoost and Random Forest offer reliable alternatives with balanced class-wise performance.

The confusion matrices, a crucial tool in dissecting and contrasting the performance of machine learning models for a multi-class classification problem, offer profound insights. A confusion matrix is a table that comprehensively presents the performance of a classification algorithm. In the context of the current multi-class classification problem, a 3 by 3 matrix is used. Each element in the confusion matrix holds significance. The diagonal elements represent the instances that are correctly classified, providing a measure of the algorithm's accuracy. On the other hand, the off-diagonal elements indicate misclassifications, revealing areas for improvement. In this study, the confusion matrices for machine learning techniques are displayed in Fig. 11.

Figure 11 displays the confusion matrices for six classification models, revealing how accurately each algorithm predicted the three stress level classes. In subfigure (a), Logistic Regression demonstrates strong performance with minimal misclassification, particularly in Class 2 where 65 out of 71 cases were correctly identified. In subfigure (b), the SVM model shows a high classification rate across all classes, with slightly more confusion between Class 0 and Class 1, though overall distribution remains balanced. Subfigure (c) illustrates that Gradient Boosting maintains decent performance but misclassifies 6 and 5 instances in Class 0 and Class 2

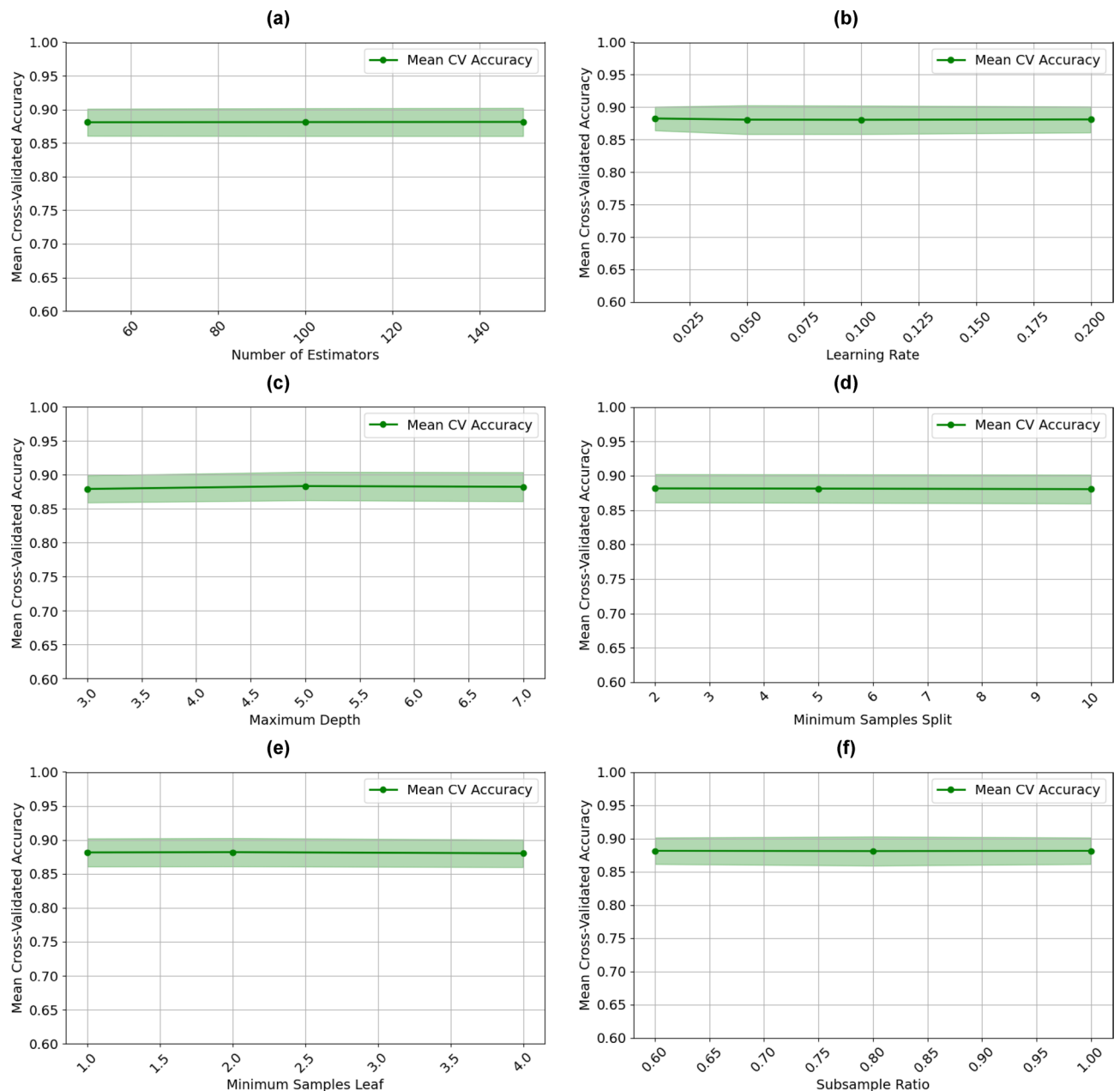


Fig. 7. Validation curves for Gradient Boosting classifier illustrating mean cross-validated accuracy across various hyperparameters: (a) number of estimators, (b) learning rate, (c) maximum tree depth, (d) minimum samples required to split, (e) minimum samples required at leaf node, and (f) subsample ratio.

respectively, indicating some loss of precision. In subfigure (d), the Decision Tree model correctly predicts most Class 0 instances (68), yet shows moderate misclassification for Class 2. Subfigure (e) shows XGBoost, which distributes errors relatively evenly and achieves high true positive counts, suggesting its effectiveness in handling class boundaries. Lastly, subfigure (f) depicts Random Forest, where the model exhibits slight confusion between Class 0 and Class 2 but still maintains solid overall classification with consistent accuracy across classes. These confusion matrices reinforce the reliability of ensemble methods and highlight the overall balanced performance of both linear and non-linear classifiers in predicting student stress levels.

Figure 12 presents a comparative visualization of feature importance derived from four tree-based machine learning models—Decision Tree (a), Random Forest (b), XGBoost (c), and Gradient Boosting (d)—to evaluate the contribution of various predictors in classifying student stress levels. In subfigure (a), the Decision Tree model clearly emphasizes the dominant influence of *sleep_quality* and *teacher_student_relationship*, which together account for over 70% of the total importance, while the rest of the features contribute marginally, indicating a strong reliance on a narrow subset of predictors. In contrast, the Random Forest model in subfigure (b) distributes the importance across a wider range of variables, with *blood_pressure*, *extracurricular_activities*, *self_esteem*, and *sleep_quality* emerging as the top contributors, reflecting a more balanced and ensemble-driven

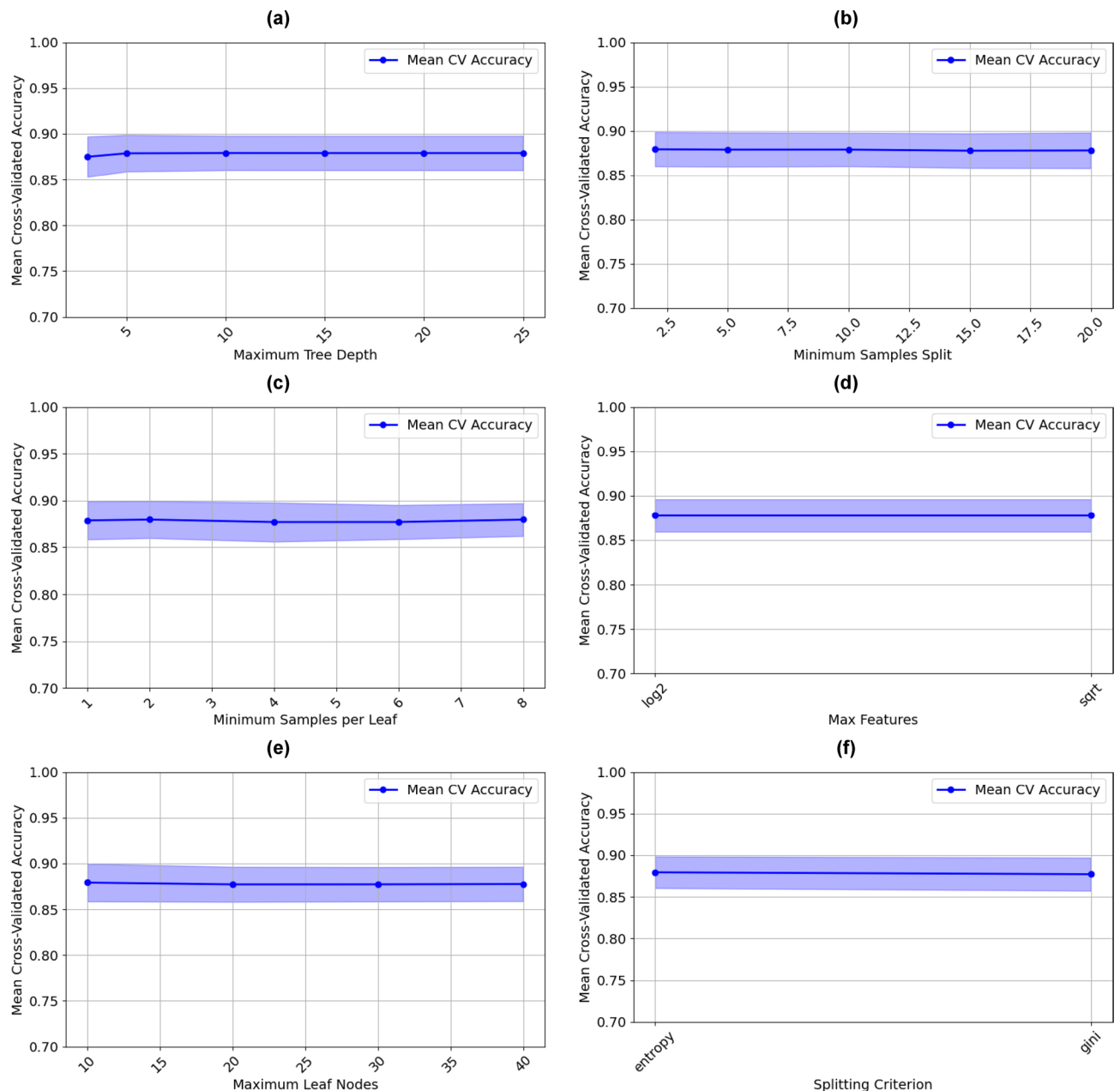


Fig. 8. Validation curves for Decision Tree classifier showing mean cross-validated accuracy across hyperparameter values: (a) maximum tree depth, (b) minimum samples required to split, (c) minimum samples per leaf, (d) maximum number of features considered at split, (e) maximum number of leaf nodes, and (f) splitting criterion.

assessment of variable influence. Subfigure (c) shows the output of the XGBoost classifier, where *blood_pressure* dominates the importance ranking, followed by *safety*, *extracurricular_activities*, and *sleep_quality*, highlighting a shift toward physiological and contextual indicators. The broader spread of importance values in XGBoost, with meaningful contributions from over a dozen variables, suggests its higher sensitivity to complex feature interactions. In subfigure (d), the Gradient Boosting model similarly identifies *blood_pressure*, *safety*, and *extracurricular_activities* as leading predictors, but also gives considerable weight to *sleep_quality* and *social_support*, confirming the multidimensionality of stress predictors spanning physical, social, and environmental factors. Across all models, features such as *depression*, *anxiety_level*, and *peer_pressure* receive relatively low importance scores, suggesting that while they are often associated with stress in literature, their statistical influence may be overshadowed by more quantifiable or consistently reported features in this dataset. The convergence of certain features—especially *blood_pressure* and *sleep_quality*—across multiple models, reinforces their robustness as key predictors. These findings support the premise that integrating physiological, academic, and relational factors yields a comprehensive understanding of stress in student populations, offering valuable insights for developing targeted educational interventions.

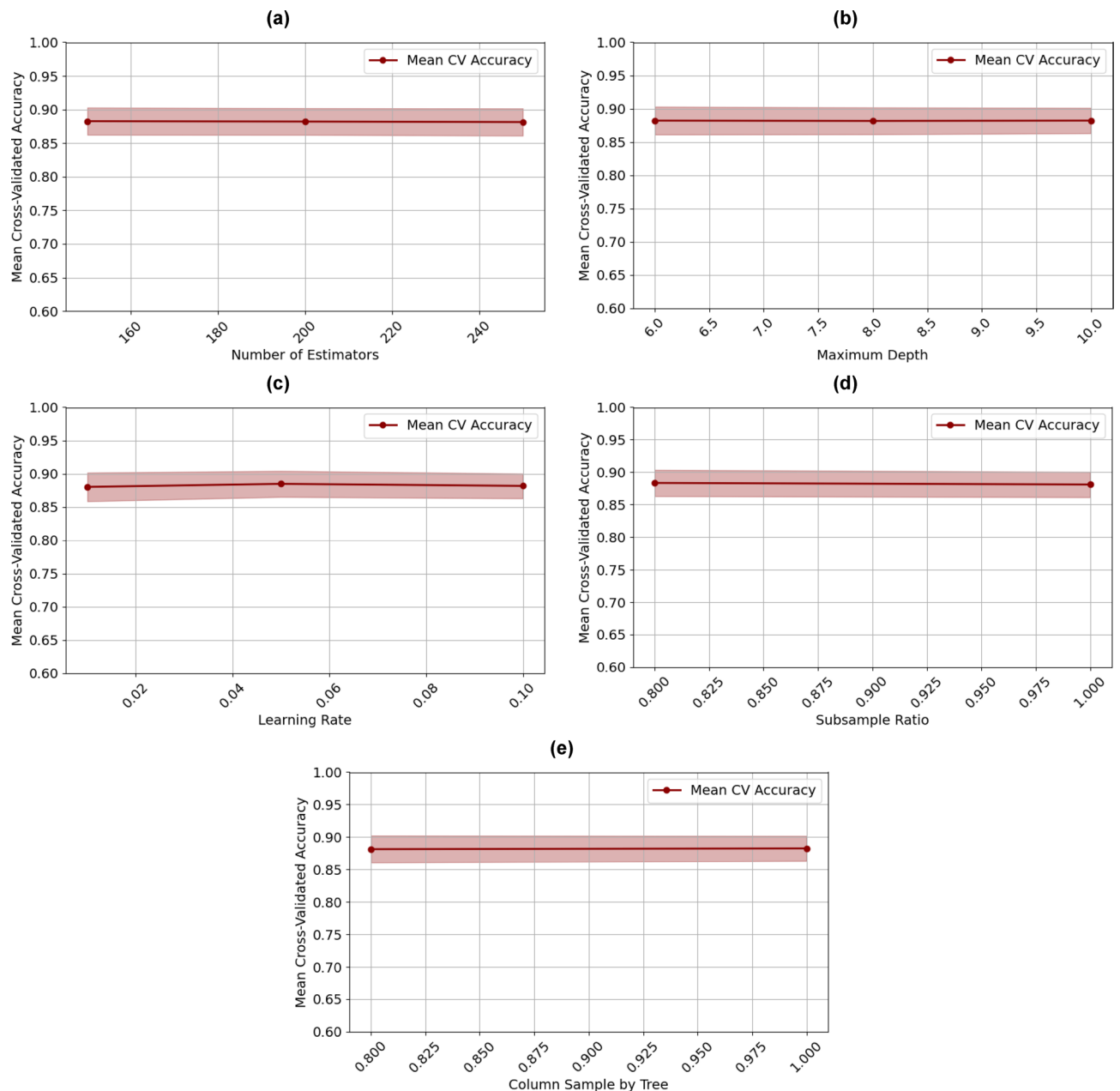


Fig. 9. Validation curves for XGBoost classifier illustrating the mean cross-validated accuracy across key hyperparameters: (a) number of estimators, (b) maximum tree depth, (c) learning rate, (d) subsample ratio, and (e) column sample by tree.

Discussion

The performance of the machine learning models highlights the role of specific stress factors such as blood pressure, safety, sleep quality, teacher-student relationship, and extracurricular activities (Fig. 12), which are consistent with previous studies that have identified environmental and physiological factors as significant contributors to student stress and also showcases the predicting capabilities of these models.

The findings highlight the complex interplay of these factors in shaping students' mental health and academic performance. High blood pressure and poor sleep quality, both physiological markers, have been consistently linked to heightened stress levels in students, aligning with prior studies that utilized advanced methods such as EEG and ECG analysis to validate these relationships³⁹. Blood pressure demonstrated the highest SHAP value among all physiological features, indicating a strong association with elevated stress levels. Ribeiro Marins et al.⁸⁸ established that lesions in the insular cortex disrupt autonomic regulation, resulting in increased heart rate and altered sympathetic nervous activity. These neurophysiological changes, while studied in animal models, suggest that chronic stress may elevate cardiovascular risk by dysregulating homeostatic mechanisms. The identification of blood pressure as a key stress marker outlines the physiological burden experienced by students under

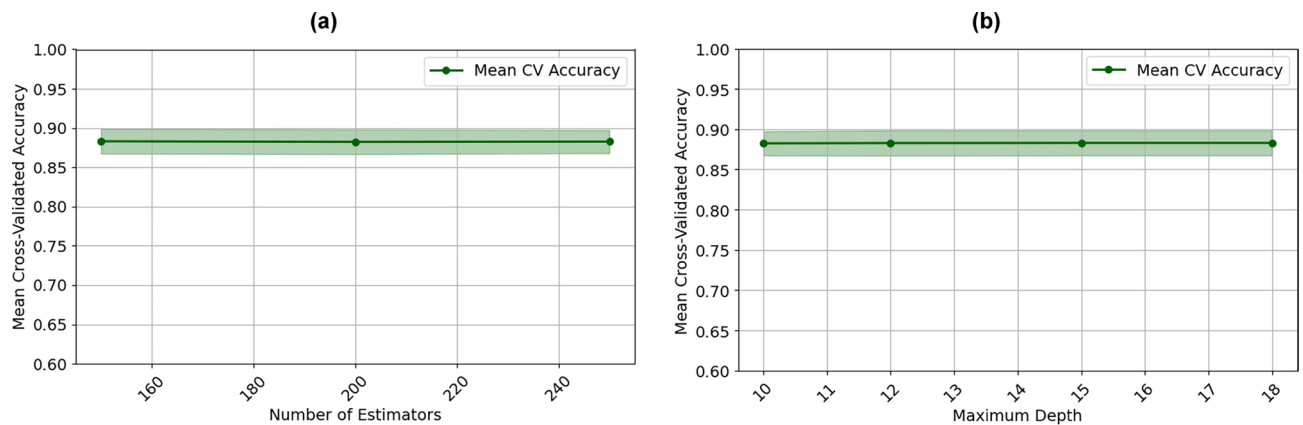


Fig. 10. Validation curves for random forest classifier showing mean cross-validated accuracy across: (a) number of estimators, (b) maximum depth, (c) minimum samples split, (d) minimum samples leaf, and (e) maximum number of features considered at each split.

sustained academic and environmental pressure. Safety, including perceptions of physical and psychological security, emerged as a significant environmental determinant of stress. Liu et al.⁸⁹ reported that children's health and well-being are influenced by community safety, with insecure environments contributing to psychological distress. In academic contexts, feelings of unsafety—whether due to bullying, institutional instability, or social exclusion—can activate stress pathways, reducing students' cognitive engagement and emotional regulation. These findings affirm the importance of designing secure and inclusive learning environments as part of institutional stress-reduction strategies. Sleep quality has a strong negative relationship with stress (Fig. 4). Kalmbach et al.⁹⁰ found that poor sleep not only follows emotional disturbances but also predicts diminished affect regulation the following day. Students experiencing inadequate sleep often display heightened reactivity to academic stressors, leading to a vicious cycle of cognitive fatigue and emotional exhaustion. The current findings validate the relevance of sleep as a modifiable behavioral factor and support the integration of sleep interventions within student support frameworks. The teacher-student relationship ranked among the top psychosocial predictors of stress. Herman et al.⁹¹ demonstrated that high teacher stress and burnout correlate with negative student behavioral and academic outcomes. In contrast, positive relational dynamics between teachers and students provide emotional scaffolding, enhance perceived support, and buffer the psychological effects of academic overload. Educational institutions that invest in relational pedagogy and faculty-student mentorship contribute significantly to lowering student stress. Extracurricular activities also showed a protective effect against stress. Fredricks and Eccles⁹² reported that structured participation in school clubs and prosocial contexts correlates with improved psychological adjustment and academic success. Engagement in non-academic activities enables students to cultivate social networks, develop coping skills, and create meaningful experiences outside the classroom. These findings suggest that academic institutions can alleviate stress by promoting access to diverse and inclusive extracurricular programming.

On the other hand, the complex relationship between student stress, higher education, and educational innovation has been documented extensively in recent research. University students across diverse cultural contexts consistently report elevated levels of stress, anxiety, and psychological burden^{93,94}. This stress is often exacerbated by academic workload, displacement from familiar environments, and transitions in pedagogical modalities, particularly during events such as the COVID-19 pandemic^{95,96}. Stress in higher education is not solely a product of academic demands; it also arises from broader psychosocial and institutional conditions. Students experience emotional dissonance, isolation, and identity-based marginalization, especially among those from underrepresented or international backgrounds^{97,98}. In such contexts, Social and Emotional Learning (SEL) frameworks offer structured strategies to buffer students against psychosocial stressors. These frameworks conceptualize support as multilayered—akin to a “Swiss cheese model”—that allows institutions to align pedagogical, emotional, and social scaffolding to support mental wellbeing in inclusive ways⁹⁷. From a health intervention standpoint, integrative strategies have emerged as valuable adjuncts. For instance, nutritional supplementation—such as B-complex vitamins—was shown to alleviate stress symptoms in tertiary students, with Standard-B formulations outperforming Activated-B in reducing self-reported stress levels over a three-week trial⁹⁹. Such findings outline the importance of multidimensional approaches that consider both physiological and psychological pathways to stress regulation. Educational innovation plays a significant role in mitigating academic stress. A seminal study using audiovisual interventions in anatomy instruction demonstrated that the integration of real-life dissection videos significantly decreased students' anticipatory anxiety¹⁰⁰. These results validate the use of innovative pedagogical tools to acclimate students to stressful learning environments. Similarly, curriculum innovation has demonstrated effectiveness in moderating academic burnout and enhancing learning engagement in secondary and postsecondary institutions^{101,102}. Furthermore, the adaptation of rational emotive behavioral therapy (REBT) in Nigerian universities showed measurable reductions in stress levels, highlighting the efficacy of structured, psychologically grounded interventions¹⁰³. Such cognitive-behavioral models, when embedded into educational innovation strategies, can provide students with personalized coping mechanisms

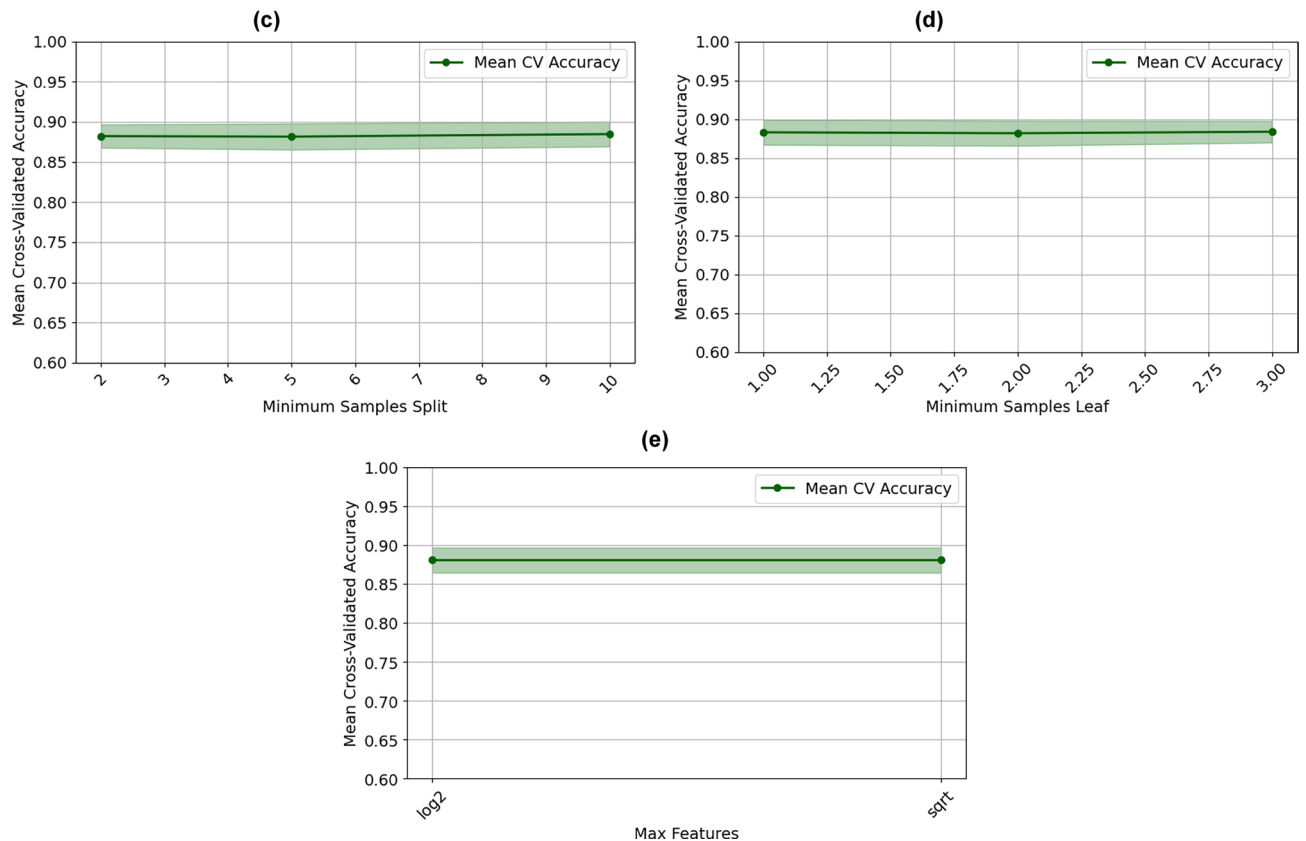


Fig. 10. (continued)

and enhance their resilience. Finally, technological transitions—especially the forced adoption of online learning—have introduced new stressors related to digital infrastructure, reduced social interaction, and altered motivational patterns. Research by Attarabeen et al.¹⁰⁴ and Chandra¹⁰⁵ found that while many students were able to adapt through emotional intelligence and self-directed strategies, the digital divide and lack of preparedness in remote learning exacerbated stress for others. As illustrated in Fig. 13, the overlapping domains of student stress, educational innovation, and higher education systems converge to form what is conceptualized as a transformative wellbeing ecosystem. This integrative space highlights the necessity of coordinated institutional action, where innovations in pedagogy and student services are embedded within systemic policies to address the multifactorial nature of student stress. By visualizing these interdependencies, Fig. 13 outlines the imperative for higher education institutions to transition from isolated interventions to cohesive strategies that sustain student wellbeing across academic, technological, and governance dimensions. Meanwhile, from an educational psychology perspective, these findings emphasize the need for specific interventions targeting specific stress factors. Programs that monitor health indicators, particularly blood pressure—ensure safety on campus, promote work-life balance to improve sleep quality, implement targeted interventions to enhance teacher-student relationships, and encourage participation in extracurricular activities can effectively reduce student stress. For instance, implementing psychoeducational modules that focus on coping strategies and resilience-building has shown significant improvements in stress management and academic performance⁵³. Furthermore, the inclusion of advanced monitoring technologies, such as wearable devices and mobile applications, provides opportunities for real-time stress detection and intervention, as evidenced by research utilizing mobile and wearable data for stress monitoring^{51,52}. There is another perspective to these findings as well because SHAP interaction values also revealed that the combined effect of certain features magnifies stress risk beyond their individual contributions. For example, elevated blood pressure predicted higher stress levels more strongly when paired with low sleep quality or weak teacher-student relationships. Similarly, perceptions of campus unsafety showed a stronger association with stress among students who lacked engagement in extracurricular activities. These interactions suggest that isolated interventions may be insufficient, and that integrated strategies targeting multiple domains simultaneously are more likely to yield meaningful reductions in student stress. Together, these findings present a compelling case for a convergent model of educational innovation that integrates technological, psychological, and curricular design elements. Such an approach is essential not only for enhancing learning outcomes but also for cultivating a sustainable educational environment that recognizes and proactively addresses the multidimensional nature of student stress in higher education.

	Precision	Recall	F1-score	Support
(a) Logistic regression				
Class 0	0.86	0.87	0.86	76
Class 1	0.89	0.88	0.88	73
Class 2	0.92	0.92	0.92	71
Macro average	0.89	0.89	0.89	220
Weighted average	0.89	0.89	0.89	220
Accuracy: 0.89				
(b) Support vector machine				
Class 0	0.88	0.88	0.88	76
Class 1	0.88	0.9	0.89	73
Class 2	0.91	0.89	0.9	71
Macro average	0.89	0.89	0.89	220
Weighted average	0.89	0.89	0.89	220
Accuracy: 0.89				
(c) Gradient boosting				
Class 0	0.87	0.86	0.86	76
Class 1	0.88	0.88	0.88	73
Class 2	0.86	0.87	0.87	71
Macro average	0.87	0.87	0.87	220
Weighted Average	0.87	0.87	0.87	220
Accuracy: 0.87				
(d) Decision trees				
Class 0	0.86	0.89	0.88	76
Class 1	0.85	0.86	0.86	73
Class 2	0.9	0.85	0.87	71
Macro average	0.87	0.87	0.87	220
Weighted average	0.87	0.87	0.87	220
Accuracy: 0.87				
(e) XGBoost				
Class 0	0.88	0.87	0.87	76
Class 1	0.89	0.89	0.89	73
Class 2	0.88	0.89	0.88	71
Macro average	0.88	0.88	0.88	220
Weighted average	0.88	0.88	0.88	220
Accuracy: 0.88				
(f) Random forest				
Class 0	0.83	0.88	0.85	76
Class 1	0.93	0.88	0.9	73
Class 2	0.9	0.89	0.89	71
Macro average	0.88	0.88	0.88	220
Weighted average	0.88	0.88	0.88	220
Accuracy: 0.88				

Table 4. Classification reports summarizing precision, recall, F1-score, and overall accuracy for each model: (a) logistic Regression, (b) support vector Machine, (c) gradient Boosting, (d) decision Tree, (e) XGBoost, and (f) random Forest, showing comparative performance across the three stress level classes.

Conclusion

This study identified blood pressure, perceived safety, sleep quality, teacher-student relationship, and extracurricular participation as the most influential predictors of student stress, as revealed through explainable machine learning techniques. The analysis confirmed that physiological, behavioral, and psychosocial domains collectively shape stress outcomes in higher education settings. Elevated blood pressure reflected physiological dysregulation associated with chronic stress exposure. Perceived safety influenced psychological stability, while sleep quality affected emotional resilience. Teacher-student interactions played a central role in academic support, and engagement in extracurricular activities provided essential outlets for social connection and stress relief. These findings support the development of integrated institutional strategies that monitor health indicators, ensure safe environments, enhance relational pedagogy, and promote balanced student engagement. Therefore,

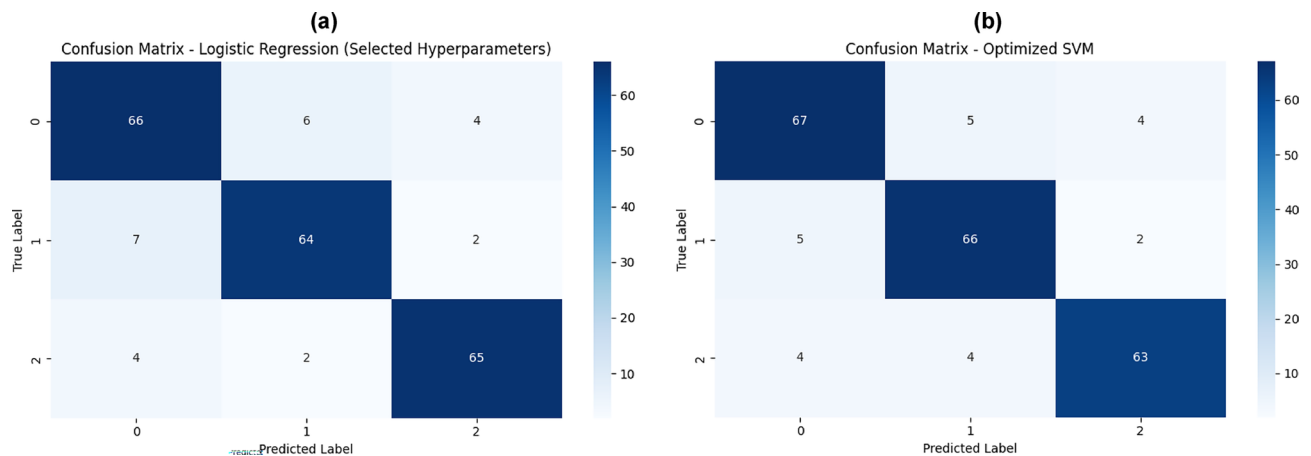


Fig. 11. Confusion matrices for six optimized classifiers applied to stress level prediction: **(a)** Logistic Regression, **(b)** Support Vector Machine (SVM), **(c)** Gradient Boosting, **(d)** Decision Tree, **(e)** XGBoost, and **(f)** Random Forest, highlighting classification performance across three stress categories.

the implication of this study implies addressing student stress requires multi-perspective interventions grounded in both data-driven insights and educational practice.

While insightful, this study has limitations. The dataset's broad scope may miss key variables influencing student stress, necessitating further exploration. Relying on retrospective data and potential algorithmic refinements are noteworthy constraints. The study's demographic focus raises concerns about generalizability, urging future research to include diverse participants and explore additional factors. Despite these limitations, the study lays the groundwork for enhancing our understanding of student stress and improving support mechanisms in education. A key limitation of this study lies in its reliance on a single dataset, which may restrict the generalizability and universality of the findings. While the dataset includes multiple variables, its representativeness across diverse educational and cultural contexts remains a concern. To strengthen the reliability and validity of future research, it is essential to diversify datasets by including samples from various cultural and educational backgrounds. Cross-cultural studies and the inclusion of datasets with broader demographic and contextual coverage could significantly enhance the robustness and applicability of the conclusions, ensuring that interventions and models address the complex nature of student stress across different environments. The study acknowledges two potential approaches to address the limitations of relying on a single dataset for stress prediction. The first approach involves developing a universal ML model capable of accommodating diverse cultural and educational contexts. However, this method faces significant challenges, as the variability in input data—arising from differences in institutional resources, regional priorities, and ethical requirements—makes such a model overly complex and potentially unreliable. For instance, institutions in different regions may lack the infrastructure or funding to collect specific data, leading to incomplete or inaccurate predictions in a generalized model. Alternatively, the second approach, which the authors favor, advocates for decentralized and case-specific ML models. These models are specific to the unique cultural, regional, and institutional contexts, ensuring that predictions are accurate and ethically compliant. By focusing on localized datasets, this approach avoids the pitfalls of imposing external factors that may not align with the specific conditions of a region or institution. Furthermore, it respects diverse ethical frameworks, as certain variables permissible in one region may require stringent approvals in others. Therefore, the study prioritizes context-specific models over generalization to maintain accuracy, ethical integrity, and practical relevance. Additionally, this study also recognizes several other important limitations that influence the interpretation and applicability of its findings. The predictive models operate under specific assumptions and may reflect algorithmic biases shaped by the training data, which affects their generalizability beyond the current sample. Correlational analyses limit the ability to draw causal conclusions, as the observed relationships between variables do not confirm direct effects. The dataset reflects a particular cultural and institutional context, which may introduce demographic bias and reduce the relevance of outcomes in more diverse or global educational settings. Additionally, the use of institution-specific modeling approaches presents ethical concerns, such as inconsistent stress metrics across regions and the risk of deploying biased algorithms in environments with limited resources or infrastructure. These constraints emphasize the need for cautious interpretation and future cross-contextual validation.

Future studies should aim to extend the scope of research on student stress by exploring longitudinal data, i.e. tracking stress factors over time, providing deeper insights into the temporal dynamics of stress development. It would also be beneficial to examine the effectiveness of different machine learning models that might offer improved predictive accuracy or new perspectives on stress factor interactions, although sacrificing transparency for performance. Furthermore, incorporating emerging technologies such as IoT devices for real-time stress monitoring and AI-driven interventions could impact the way educational institutions manage and mitigate stress. Such advancements could refine the predictive models while enhancing real-world applicability, allowing for the development of personalized student support systems. Future work will focus on conducting

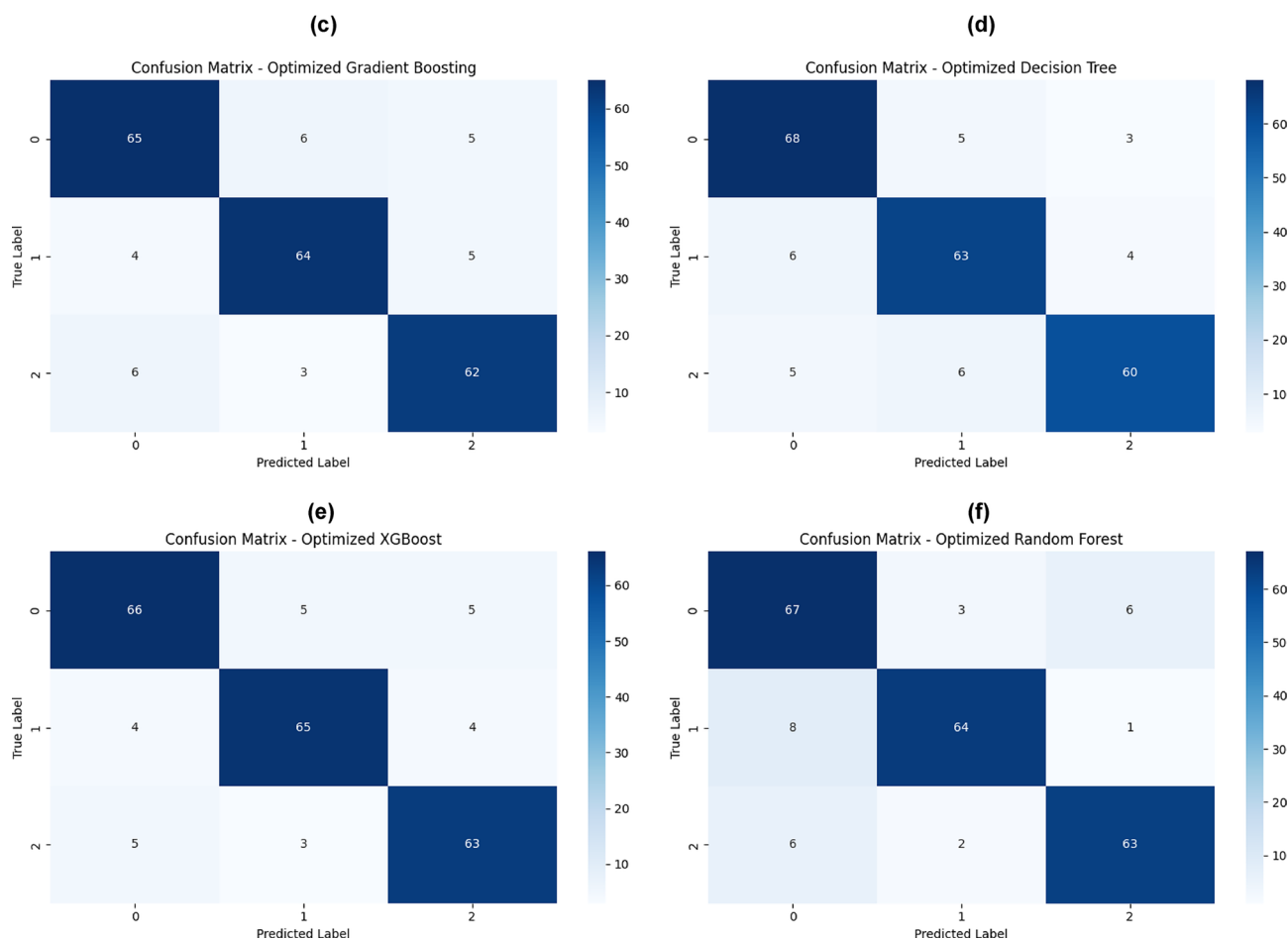


Fig. 11. (continued)

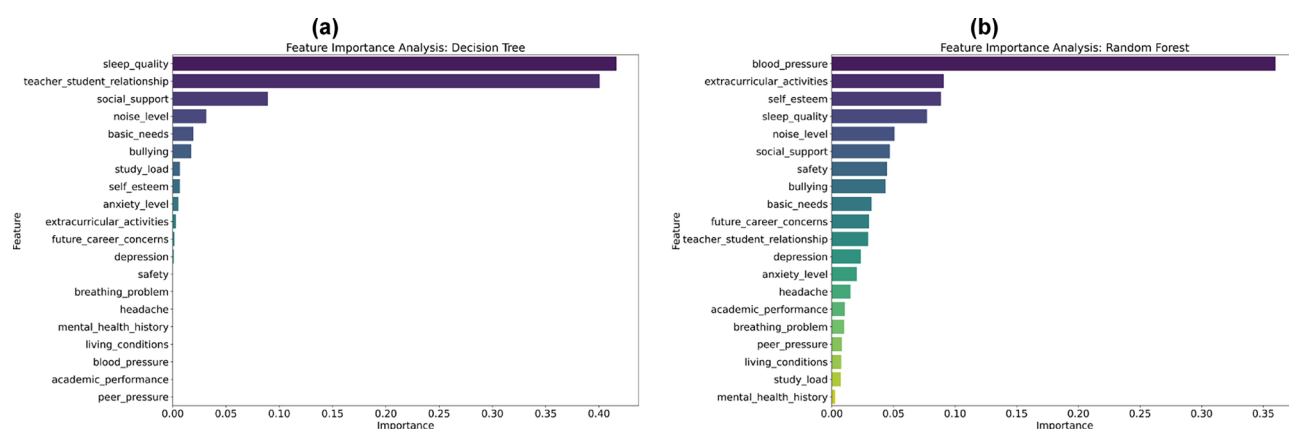


Fig. 12. Comparative feature importance plots generated from four machine learning models: **(a)** Decision Tree, **(b)** Random Forest, **(c)** XGBoost Classifier, and **(d)** Gradient Boosting, illustrating the varying influence of predictors on stress level classification.

comprehensive experimentation to address potential overlaps among variables, such as the influence of other factors on blood pressure, building upon this study as a foundational methodological framework.

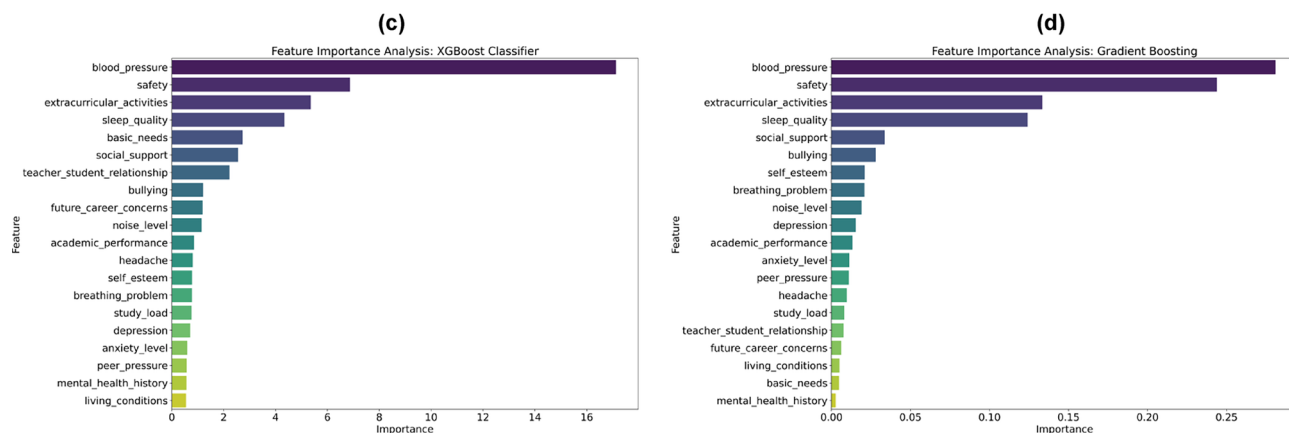
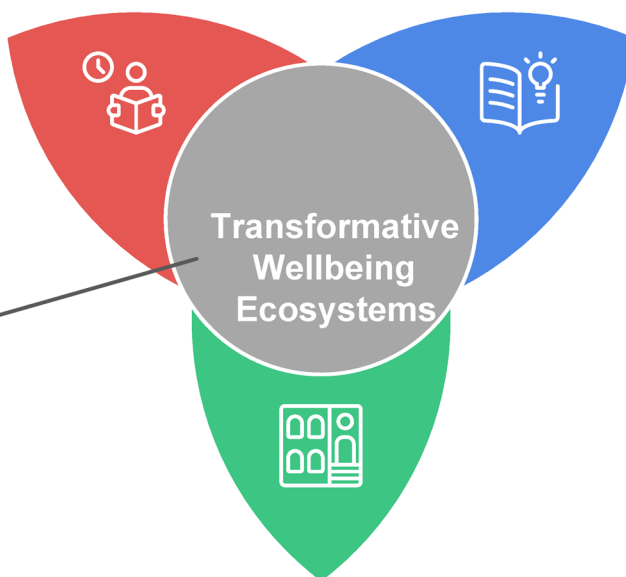


Fig. 12. (continued)

Student Stress

Refers to the psychological, emotional, and physiological burdens that affect students' academic performance and personal wellbeing.

Promotes a unified approach to student wellbeing by aligning institutional policies, pedagogical innovations, and psychosocial support to address academic stress within higher education.



Educational Innovation

Involves the integration of novel pedagogical strategies, technologies, and support mechanisms to enhance learning and reduce academic strain.

Higher Education Systems

Comprise the institutional frameworks, policies, and governance structures that shape the academic and social environments for students.

Fig. 13. Conceptual model illustrates the intersection of student stress, educational innovation, and higher education systems, emphasizing the need for transformative wellbeing ecosystems to promote integrated and resilient academic environments.

Data availability

The complete code is available on the following link: [https://github.com/RasikhTariq/machinelearning/blob/main/Article2_StudentStressLevel_\(1\).ipynb](https://github.com/RasikhTariq/machinelearning/blob/main/Article2_StudentStressLevel_(1).ipynb).

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Author contributions

Rasikh Tariq did all the data analysis and wrote the results. Muhammad Tayyab Zamir developed the method section. M.G. Orozco-del-Castillo developed the overall flow of the manuscript, developed the problem statement and the discussion. , Maria Soledad Ramirez-Montoya wrote the theoretical framework. Tabbi Wilberforce bought coherence in all the sections, wrote some parts of the manuscripts, and prepared the final draft.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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